U-Mentalism Under the Radar of Patent Law Great Peril

"What we're dealing with here is a total lack of respect for the law"

Their Law

Prodigy, Music for the Jilted Generation (1994)

Abstract:

Amidst the recent legal advancements surrounding the Unitary Patent, this paper aims to highlight a critical risk inherent in patent law—one that extends beyond a mere legal loophole to a potentially hazardous flaw that threatens the very foundation of global patent protection. This risk, deeply detrimental to innovation and inventive human endeavour, is rooted in the criteria for patentability as outlined in WIPO's Rule 39 PCT Subject-Matter, Article 17 (2) (a) (i). Specifically, this rule states that: "*No International Searching Authority shall be required to search an international application if, and to the extent to which, its subject matter is any of...*" Among the six specified categories, the one most relevant to this analysis is: *"(i) scientific and mathematical theories.*" The case study at the center of this discussion is a novel computer architecture and supercomputing model known as U-Mentalism (PCT/PT2021/050014), for which the author is also the inventor. This paper will also examine the apparent merits of historical American patent law, particularly in contrast to the European Patent Office (EPO). The analysis will emphasize the substantive implications of Rule 39's exclusions and argue against allowing *Search Authorities* (SAs) to be exempt from issuing *Search Reports* (SRs). The overarching recommendation is a legislative shift that mandates *Search Authorities* to conduct and provide *Search Reports* under all circumstances, ensuring a rigorous framework of testable, falsifiable oversight and control case within the critical limits of philosophy of science.

Philosophy of Science: the demarcation question

U-Mentalism (U.M. for short), formally titled **"***Method to Perform Computation at or Near the Speed of Light (Typical) Digital Image-to-Binary Singular or Multiple-Nodes/Servers and Computer Architecture***"** (PCT/PT2021/050014), introduces a new computer architecture model. This model replaces binary code with an isomorphic RGB colour code (or rather, in physical terms, wave-packets of light), thereby enabling computation per frames cinematically, and wherein locally at each pixel, inasmuch as per each frame, processing is performed both orthogonally and synchronously.

After its initial submission (earliest priority date: May 20, 2020), the European Patent Office (EPO), following norms under the Patent Cooperation Treaty (PCT), issued a "*Declaration of Non-Establishment of International Search Report*" under "*PCT Article 17(2)(a), Rules 13ter.1(c), and Rule 39*". The primary argument cited was that the invention fell under the category of **"***scientific theories***"**. Strangely, EPO's declaration also claimed that both the description and the claims failed to meet prescribed requirements (mailing date: August 24, 2021).

Fortunately, WIPO's legal framework allows for provisional patents and mandatory national-level evaluation of Intellectual Property (IP) submissions, which is what inevitably befell. Consequently, the case was automatically forwarded to national and regional offices for further assessment.

It is the understanding of the inventor and author that one such "*Declaration"* (of non-establishment of international Search Report) necessarily translates to a major red flag alert, to the extent of representing a *casus belli* of property rights protection on the international patent arena, worth being signalled from the very start under a vigilant watch (to the extent of a permanent report under patent law journals or academic publications), for neither the theoretical concepts, neither the technical basics, or even juridical key fundamentals were demonstrated to be mastered by the body of EPO examiners.

The lack of mastery over theoretical, technical, and juridical fundamentals in this case and on the whole draws EPO's demandable expertise away from a natural *sui juris* autonomous stance, instead collapsing onto the verge of a fatal *de jure* flaw. Astonishingly, there are appreciation errors even in relation to state-of-the-art searchable concepts found in Wikipedia, such as the "pixel" and its correlatives, as one reads, for instance, "*Moreover, the term «pixel equation» is neither a term known in the art, nor is it defined in the claim*" (PCT/PT/2021/050014). To make it worse, there are similar takes on related concepts such as one against the use of self-explanatory universally transversally acknowledged term "cells", a term that is commonly employed across disciplines including biology, computer science, and data processing, not to mention nitpicking assaults on new best tailored concepts, an absolute need for inventions of novelty to freely equip themselves with, provided that they are thoroughly explained, which was clearly the case.

If it was only a matter of unwary and improvident matter before us (i.e., sporadic, noticeably non-methodological and non-reiterated), the overall judgment would be very far from an incontrovertible evaluation of very pernicious traits. Yet, in the author and inventor´s judgment, it is rather the non-incremental, very much to the contrary, exponential nature of the invention in conjugation with the latter that causes one such great alert. Be it as it may be, one such odd cascade of misevaluations by the body of examiners of EPO could have come from many corners: either from institutional policy bounding, possibly even self-inflicted shadowy constrains, or worse, direct cynical

deviant and unethical lobbyism, with the result of patent obstruction from vested distorting interests, namely by the largest corporations and players in the same areas of industry, research and innovation, concurrently the same largest patent holders and liquid contributors to WIPO´s different national regional offices around the globe.
More troubling is the broader issue of technological gatekeeping. If left unchecked, such flawed assessments could create a precedent for suppressing disruptive innovations, particularly those that introduce exponential technological shifts rather than incremental improvements. The U.M. case raises concerns about whether patent law is being weaponized to stifle groundbreaking innovation in favour of entrenched market players. Fundamentally, the author argues that patent authorities must be held to rigorous accountability standards. This case study highlights the perilous consequences of allowing *International Searching Authorities* (ISAs) to reject patents outright without conducting thorough "*Searches*". Given the global impact of patent law, the absence of robust evaluation mechanisms could seriously undermine the legitimacy of the entire patent system, stifling innovation at a fundamental level.
Often, the "elephant in the room" phenomenon is rendered nullified, literally *nihil*, a perfect unspoken void, unless it is equated, named, covered, and critically discussed. If not, it will come unstuck decades later, bedraggled in folk media and kitsch culture, when there is nothing left but flagrant grievous injustice and inventors' great defeats, often accompanied by massive theft. This is exemplified in pop culture, yet with sophisticated technicalities in both software and judicial pleading, by the recently popularized Netflix series "*The Billion Dollar Code*", which depicts the true event of ART+Com vs. Google, where the case was shamefully lost both at trial and on appeal (USRE44550E1 "*Method and device for pictorial representation of space-related data*").
Unquestionably, vague legal provisions such as the possibility of denying "Search Reports" create an opening for harmful intentions and at the limit technological monopolization, while it is also true that this critical vulnerability, or rather a major flaw in patent law administration & legislation, reinforces the argument beyond the need that examiners are held accountable for their assessments, and demands that a philosophy of science & technology scrutiny follows for fair and impartial evaluation, perhaps only in the most difficult cases, but definitely all having been the subject of a *Search Report* priorly.
In a world where intellectual property is increasingly valuable, the absence of mandatory production of *Search Reports* by *Search Authorities*, beyond consistent and thorough examination procedures, severely threatens the legitimacy of the patent system and also the protection of groundbreaking innovations.
Also it is not to leave behind the permitted conclusions that the U.M. control case suggests, i.e., not only the far more recommendable and refined United States Patent and Trademark Office (USPTO) patent law philosophy and history in the strict terms of patentability criteria (a remark even before the arrival of USPTO´s Report on the invention), whose explanatory technical, philosophical, and historical causes must be reflected, or better even the necessary avoidance of the great pitfall of having WIPO´s Rule 39 PCT Subject-Matter, under Article 17 (2) (a) (i), allowing *International Search Authorities* (ISA), in the form of national and regional patent offices (moreover with

associated high fees for this very same purpose), not establishing any *Search Reports* (SR) at no point and in any known circumstance. In place of such it has come into view one very convenient vague, very hazy, WIPO´s Rule 39. It is precisely on this hollow and void legalistic *stratum* – again, where *Search Authorities* (SA) do not produce *Search Reports* (SR) — that the great peril of patent theft, obstruction, and open litigation, impends most heavily. And even more so when, in abstract, there is a production of novelty that escapes the dominant marginally graded, step-by-step incremental typology of inventions, one such in place as that of a radical exponential change and vast technical drift could be.

Taken that at the heart of EPO´s Report in the case of U.M. is the assertion on the legal base of Article 17(2), Rules 13ter.1(c) and Rule 39 (PCT/PT/2021/050014), namely, that the subject matter of the international application is a "*scientific theory*", thus one such presumed cause rendering a failure of both the description and the claims to comply with prescribed requirements, preventing a meaningful *Search* from being carried out, concurrent to this paper and critical *Opposition* (O), aiming to better sustain the arguments on behalf of the petitioner and author, it is desirable to substantiate statements of open defence under the entire width and full comprehensiveness of all legal forms, general *Guidelines*, *Rules* and *Articles* — but because this is not the place for a formal "*Opposition*" to take hold versus any *Written Opinion* (WO) of any *International Searching Authority* (ISA) — it is, yet, adjusted to this paper´s scope the broadening of arguments (legalistic, philosophical, cultural, historical, etc), the first of which and most prominent a call to order on demarcation in general history and philosophy of both science & technology, under the sight of patentability criteria.

Having the author and inventor received a doctorate in logic and philosophy of science, this matter assumes thoroughly further raised relevance, mainly because the preliminary question of demarcation in science is determinant, by the effect of the latter on what falls under scientific, yet also technological patentability criteria — "*a new product or process involving an inventive step and capable of industrial application*" (Section 2(j) of the Patents Act, 1970) — furthermore in legalistic statutory cloth, mostly when this thin shifting line was, indeed, for any glance of scientific theories judgment, *ab contrario* forethought by the author and inventor in crafting the arguments and claims in the submitted patent.

Ultimately, by upfront *reductio ad absurdum argumentum*: if a presentation of a technicality with a visible assignable purpose, circumscribed as it might be under a scientific-based action, typically a process or device, is not *tout court* and *per se* a scientific theory, even if holding coinciding features with one, then it is, compulsorily, patentable subject-matter. It is warranted that postulates of any scientific theory can be an invention if promote, demonstrably, a new method or process, but established the latter and in case it is not even a scientific theory, no doubt whatsoever impends on patentability criteria, and the oddest it is when patent licensing bodies force its rejection.

Prosaically presented as it is for now, it ought to be said that this topic of interest invites the excelled state-of-the-art contributions of prominent philosophers such as Karl Popper (1902-1994) and his principle of falsifiability and testability; Imre Lakatos (1922-

1974) and his shift towards research programs appraisal and ground for substituting predictions; Thomas Khun (1922-1996) and his scientific paradigms empirically shared framework and revolutionary turns; Paul Feyerabend (1924-1994) and his anarchy in epistemology and the necessary breaking of methodological rules; David Miller (1942-present) and his probabilistic and hyper-rationalist "verisimilitude" in scientific inquiries; Laudan (1941-2022) with his "research traditions" as an alternative to Lakatos' programs, etc.

Acknowledging that this list is but a board of honour for a selection of the most emergent and transitive concepts in the field, there are many more eminent contributions of equal numbers, when not in significance — e.g., Willard Van Orman Quine (1908–2000) and his challenge to analytic-synthetic distinctions and the power force of synthetic judgments; Bas van Fraassen (1941–present) and his constructive empiricism, and adequation in place of truth; Ian Hacking (1936–present) and his emphasis on experimentation and reasoning styles — are all good examples, inasmuch as there is, beneath contemporaneity, figures of relevance that have shaped foundational interpretations in philosophy of science, such as Peirce´s (1839-1914) and Bachelard´s (1884-1962) views that are naturally omitted, when not buried, even if bestowed.

Bearing in mind that patentability criteria "*Rules and Articles*" cannot be jetted as paraphrases of philosophical mould, as common sense sharply dictates, what is relevant, nevertheless, is the basic underlying idea that, at least and minimally, philosophy of science & technology, notably on the level of the demarcation question, can decisively elucidate what is and what is not a scientific theory, even if under the strict terms of patentability criteria. Now, none of the above can be settled without sheltering argumentation guidelines for that same matter. It is, meticulously, at the head and foreground of such argumentation guidelines that, in turn, WIPO´s general guidelines do converge, being the case of U.M. made an imperative under-watch control case in both general philosophy of science & technology, and also mandatorily for patentability criteria´s fairest and most equitable theoretical framework conceptualization. Insofar "What is a scientific theory?" is telling of the demarcation question in philosophy of science, standing in opposition, for patentable subject-matter, "What is technology?" ahead, in regard to patentable subject-matter, is also telling of what is not a scientific theory. Regarding, thus, general arguments in philosophy of science at its core, some notes are fairly welcomed:

A non-falsifiable scientific theory in Popper's imprint would either translate to an overly dogmatic scientific corpus of laws or be classified as pseudo-science, so to establish the two extremes from which no theoretical surplus would be attained to patentability criteria. But it is worthwhile to pursue one such argument declaring that major boundaries delimitative plea for the demarcation under open enquiry in philosophy of science, in inverted fashion, can assume to be, again by direct contrast, discerned under patentability criteria, more so in cases of positive direct applicability by action of an order or process, i.e., conditioned by any agency, diligence, industrious materialization or some sort of *actus* embodiment, never excluding experimentation and testing. A prime case of notice for this take would be the difference between Faraday's

experimentalism and Maxwell's scientific theory of electromagnetism, irrespectively of the apparel used by both men to experiment and test, and at a later stage, technological new machines, processes and/or embodiments that make use of such in perfected manner.
More so, going easily unnoticed, is the importance of falsifiable scientific theories in articulation with some action or process, under some form of materialization or embodiment, to go past approval in the patent system, as long as the subject matter is not alone a proclamation of a scientific theory (beyond being non-obvious and characterized by novelty), being unmistakably attached by a process, machine, manufacture, or composition of matter" (WIPO). A prime case of notice for this take would be the difference between a Turing machine, a mathematical (recursion or computability theory) else said scientific theory, that is, most surely, not patentable, and, in opposition, a computer architecture or processing method, fully patentable as they recognisably are. Without the need for lodging any further arguments, it is precisely on this stance that the control case for the U.M. computer architecture is most relevant.
Now, another argument of importance in the demarcation question in philosophy of science is that of the so-called methodological naturalism. It is hardly taken, however, except for the guiding inquiry affirming that explanations should rest exclusively and avowedly on natural causes or phenomena (rejecting metaphysical entities, and any of religious coinage).
But there is also methodological naturalism ascertained against not metaphysical teleology, but technological, mechanized, computed, analytic, processual teleology or chain of causes, even if sharing the reductionist sight prototypical of hard scientific theories.
In other words, if it is not recognisable any technological process by which an invention visibly and recognisably determines a result, process, or new product, then it is tantamount to postulating a miracle, i.e., an extrapolation beyond both natural and technological phenomena. No one desires to see an invention vetoed the entrance in the patent system, and thereon demonstrably and exemplarily work, to such a degree even that surpasses the efficiency of older methods, being in this stage of nonsensical bizarrerie, dubbed a miracle. Would, really, EPO want to shamefully face a long-lasting fully universally deployed working method and/or process, under a new computer architecture with far more effective computability, without a suitable conceded patent, demeaning the proper patent office activity and history?
Indeed, in such cases, it should be possible to initiate legal proceedings with open-ended deadlines to seek compensation, as only by allowing sufficient time for the invention to be developed following its hindrance can the resulting losses be properly assessed and gauged.
As for logicizing concerns entangled with empiricist verifiability, namely the crossroads of consistency with evidence, or in general, observance correctness, to freely use an expression of philosophical hallmark, it is also fundamental to attain the conclusion that all the more a process can account for diverse data and withstand empirical challenges, that does not mean, in what regards patentability criteria, that it avows a scientific status. A prime case of notice for this take would be A. Graham Bell´s (1847-1922)

photophone (U.S. Patent 235,199 "*Apparatus for Signalling and Communicating, called Photophone*") issued in late 1880, an example of telecommunication device on the basis of light beams that made perdurable its principles until the fibre optic era as of now, or the case of the ENIAC patent (U.S. Patent 3120606A "*Electronic numerical integrator and computer*") on the general purpose electronic digital computer by John Eckert and John Mauchly, which is the basis for the modern most distributed known (in paraphrastically manner) as "von Neumann", or else, "Princeton" computer architecture. These examples are vitally crucial as they are affiliated with the control case under examen, the author and inventor envisagement going as far as considering U.M. a sort of midway technological pivot or break-even point alongside the technical orientation and scope of both.

Imre Lakatos (1922-1974) demanded recognising the importance of problem-solving abilities in the realm of scientific theories as a premise for successful prediction and generation of knowledge, something that is imminently and dangerously neighbour to the definition of inventions, and therefore close to patentability criteria.

As such, it is advisable, again, not to misunderstand the broadened scope of one invention and its strong inclination to prove effective in problem-solving, beyond even making successful long-term predictions, while remaining aligned with duty complied patentability criteria, as distinct from a scientific theory at the other extreme.

Rule-abiding physicalism, as executed by any machine or process, cannot be considered law-abiding physicalism as one finds in a scientific theory. A prime case of notice for this take would be the recent necessarily trivial, but exquisitely emerged "abstract idea" which, amid a sea of ambiguity, claims that a software-based invention is patent-eligible only if it '*improves computer functionality*', i.e., if, for instance, it enables previously unavailable or impossible to be performed methods, even if by differently superior computing devices, definitely ascending to higher targets in computational complexity and/or previously available computing processes.

Again, U.M. as a new computer architecture and super-computation method on the role of our study control case in both patentability criteria and philosophy of science demarcation question, fits well as conclusive evidence along the critical line in favour of a new method of computation, a new machinery process, hence abiding to all the patentability criteria, even if its broaden scope and massively-distributed problem-solving abilities in the mathematical and scientific realms, are no shorter than exponential, reason enough why the author and inventor felt the need to differentiate "U-Mentalism C" for "computation" and U-Mentalism "O" for "ontology" in journal publications of support to the general idea.

Thereupon, under this guise, a theory that continually makes progress in explaining new phenomena and resolving scientific problems is considered more scientific, but that role can also be performed, and extravagantly in better fashion, by an invention, with the von Neumann computer architecture being the sharpest example (U.S. Patent 3120606A). Across all scientific domains, from pure mathematics to cell biology, from non-linear equations to prime number calculations, from DNA libraries to protein synthesis, and from civil, aircraft, and naval engineering to chess and seemingly trivial pursuits such as word puzzle games, all have been perfected by the arrow from a Turing-

machine mathematical idea to the so called von Neumann computer architecture, none of which would be possible without the transposition of the former into the latter, i.e., the industrious idea that permits the application of all the forerunner conceptions into a machine or process, fully patentable as it is.
Also of relevance is the fact that a software patent correlates with computer programs, libraries, user interfaces, and even algorithms, none of which are at the core of the submitted patent control case, rather a computer architecture and process is. The control case study U.M. classification is as such: "(G0671/20 *Classification covering simulators which are concerned with the mathematics of computing the existing or anticipated conditions within the real device or system; simulators which demonstrate, by means involving computing, the function of apparatus or of a system, if no provision exists elsewhere; image data processing or generation*)".
This classification is duly truthful and at least partially accurate, even though machines characterised by their structurally interconnected functional elementary mechanisms as those prefigured by U.M. can, most definitely, account for new computer programs, libraries, interfaces and algorithms. This is probably the most heterogenous sign of recent-history, yet, resurgent, patentability quarrels and need of historical revisions, as it is the place whereupon the set of standards, rules and guidelines that determine whether an invention is eligible for a patent are most untoward and dreadful. Patent law interpretation is, we discover, the proper examination of patent applications. However, it is not, in any way, concealed for the skilled in the art the rationale of software patents at the core, especially for those who have developed their expertise on an executive body as large as the European Patent Office (EPO) who ought to carefully study patent law inasmuch as patent applications. However true, the latter is less of a fault compared with the alleged status of a *'scientific theory*'", a gross mistake with unpredictable consequences, what is more not even bearing the outputting of a *Search Report* (SR), hence a pitiful extrication of absurd ideas into one brief paragraph note. It certainly is not a matter only of misclassification, but the gravest and most troubling error with unforeseeable repercussions of permitting classification without the issuance of a *Search Report* (SR), not anymore reducing complex and significant considerations to a brief and insufficiently justified paragraph, but instead to an unfathomable void.
This peril is about as big as the patent system at its scale, as the author and inventor analyses it, to the point where, at this moment in time, most probably a scientific body of consultants, or even maybe a permanent scientific committee trained in philosophy of science & technology needs to furnish WIPO´s constituency with enough agential delegation so to reach national regional offices and boards of appeals with accrued competence and epistemological subject mastery, to prevent this and other cases .
The establishment of such body would hopefully ensure that national and regional patent offices, as well as boards of appeal, would operate with enhanced competence, thereby preventing this and similar cases in the present and future to come cutting-edge technological age.
It can also be foreseen from where the widespread reception of debates has arisen, such as those concerning reproducibility and transparency in the philosophy of science.

Mainly, the idea of independently verified scientific enquiries is of extraordinary importance, extending the argument additionally from the demarcation question to patentability criteria, as it is highly undesirable for major institutions to experience unfortunate mishaps and practical setbacks or vicissitudes — not only within *Search Reports* (SR) but, more critically, in the broader legal framework that permits *Search Authorities* (SA) to entirely forgo issuing a SR. This occurs under the flawed pretext of classifying an invention as a '*scientific theory*', a rationale that serves as the perfect legalistic loophole, akin to an alibi in criminal law, when misapplied in the context of patent law— potentially at the expense of exponential and otherwise highly valuable inventions, ideally preventable by some.
A prime case of notice for this take should be EPO’s own mission, values and corporate policy, not just for patent examiners, who operate at the forefront of technology and likely push themselves strenuously to extreme limits in handling the latest and most complex technical innovations, but also for procurements departments, boards of appeal at its height, and even the production of statistics and reports.
The fundamental basis for the importance of independently verified and autonomous legal and procedural examination is rooted in key legal texts, namely the *European Patent Convention* (EPC) and the *Patent Cooperation Treaty* (PCT). Without adherence to these frameworks, compliance integrity, international patent protection, and the credibility of patent office granting decisions risk being severely undermined—leading to a deep loss of trust, the erosion of shared technical knowledge, and even the premature forfeiture of public access to crucial information.
Prolonging still arguments in correlation with philosophy of science, it should be acquainted also parallel principles such as Occam´s razor parsimony, for the preferable simplicity equivalent in patentability criteria, as long as there is a revindication and/or set of claims that calls for a process, a manufacture, etc, thus exposing the minimal strong evidence in favour of technicalities use and/or mechanic powers, which conducts automatically to patent granting be its filling error free, insofar this is precisely what patentability criteria, through WIPO, working as the global forum in public international law for intellectual property, sustains in its body of law. If the world of commerce, enterprises and industry does not see the utility in exploiting that same method or technicality, that is where and when it is lost, at least in the time scope of twenty years, a pity to some of the best inventions, as the best they are the most anticipative they are too, a rationale that drives us to other contentious questions in patent law that, although linked to the U.M. case, are not the subject-matter for this article.
At last, we arrive at one of the most significant concepts in the philosophy of science: consensus as widespread acceptance. In the context of patentability criteria, this is understood as an arc from *Logos* to *Technē*, conferring legitimacy both to scientific theories and to patent granting. A prime case of notice for this take is represented, in the author and inventor´s choice, precisely by scientific journals — including patent law journals and books — as they permit the elevation of control case studies to semi-public/public exposure and scrutiny. Most probably in rarity of numbers and substance, but at least ideally, these academic and independent journals and books single out a

crucial outlet for the otherwise narrowly specialized field of patent law practice, even more so amidst the fog of millions of patents submitted each year worldwide.

Often forgotten in philosophy of science postulating drive is, most surely, the value of scepticism. In terms of patentability criteria, a sane culture of scientifical scepticism is as valued as required. Nevertheless, it cannot be twisted to the point of disowning sound arguments, so that the state of play of one industry can survive, burying both the idea of science and the scientific method at one stroke. If such a scenario occurs, then it becomes acceptable, under the guise of rational scepticism, to invoke the classic detective story trope where the least likely character turns out to be the culprit—the actual murderer: *'The Butler did it!'* Alternatively, folk wisdom expresses a similar sentiment: *'No creo en brujas, pero haberlas, haylas'—'I don't believe in witches, but indeed they exist.'* This peculiar abductive reasoning suddenly becomes the paraphrastic equivalent of the EPO's unusual rejection of a *Search Report* (SR). To close this section and prepare ahead the next that shifts more narrowly towards patent law, let us polish, in the transition, some important topics for the control case study of U.M.:

According to Article 52(2)(a) of the EPC, *'1) European patents shall be granted for any inventions, in all fields of technology, provided that they are new, involve an inventive step, and are susceptible to industrial application.'*, hence explicitly establishing that the U.M. invention fully meets these requirements.

Consequently, it must be subjected to semi-public or public scrutiny as a control case study in patent law—particularly in relation to patentability criteria. This scrutiny is crucial in assessing the risks and perils posed by the execution of examiners' *Written Opinions* (WO), which, in effect, allow *Search Authorities* (SA) to refrain from ever producing *Search Reports* (SR) for inventions or new methods.

All the stranger and puzzling is the fact that, having first been submitted to the national office of its country of origin, the corresponding national patent office initially approved it smoothly, without objection. However, in a striking display of how subserviently uncritical and lacking in autonomy a national institute can sometimes be—particularly amid a deficient historical, legalistic, scientific, and technological culture—INPI (the *Portuguese Institute of Industrial Property*) initially issued the inventor a sanitized, hypocritically concealed, and conveniently whitewashed report. This report was copied *ipsis verbis* from the previous EPO notification, contradicting INPI's own initial report. Yet, upon reassessing the case more independently and autonomously, INPI has now rightly granted the U.M. patent once again. The reason why it did not decide to do that firsthand is answered by the blind proclivity to follow EPO, as if it were the leading church of patent offices, where descending shared liturgies of *Report* writing from regional context is not precluded.

This, too, is a serious mistake, bordering on absurdity, as for most cases, inventors with a tight budget, no grants or support, furthermore not forewarned of these institutional mimetics, can easily let a patent fall due to the high fees. No one is, really, willing to pay thousands for a *copy pasted* report, especially when the very same institutes responsible for enforcing anti-counterfeiting regulations so readily engage in replication when it comes to report writing.

As we read in *Exclusions from Patentability: How Far Has the European Patent Office Eroded Boundaries?* (Sigrid Sterckx; Julian Cockbain; Cambridge University Press, 2012), with regard to 'scientific theories' and the altered meaning of 'discovery'—in contrast, for instance, to the term's original imprint in Article 1, Section 8 of the U.S. Constitution: *'In current usage, however, "discovery" normally means finding something that pre-existed, with properties that were already in effect at the time of discovery. One discovers a previously unknown plant or mineral, or that energy is proportional to mass squared, or that the square of the hypotenuse equals the sum of the squares of the other two sides of a right-angled triangle, or indeed that a gene sequence codes for a protein.'* — none of the latter prefigures the sense contained in the author and petitioner´s invention and claims, yet categorized by EPO examiners as a *'scientific discovery*'.
Although Article 52(2) of the EPC was not explicitly cited by the EPO body of examiners, its implication is evident— particularly under the letter of the law:
'The following, in particular, shall not be regarded as inventions within the meaning of paragraph 1: (a) discoveries, scientific theories, and mathematical methods.'
This exclusion was explicitly referenced in the *Declaration of Non-Establishment of an International Search Report*, which was officially communicated to the author.
In all relevance, the U.M. invention does not merely identify a new method, but mainly settles, through the steps in consonance with the claims, a new method΄s universal applicability to whichever type of computer, technical apparel or, in short, any Turing-machine (including Universal Turing-machines), as an integrally new method that, by its architecture, processes through computability.
It would be astonishing to have received from the *International Searching Authority* (ISA) office of examiners a judgement, based on from *Patent Cooperation Treaty* (PCT) *Regulations* and *Articles*, stating that the invention claims were pre-existing, the properties of which already taking place at the time of the disclosure of the presumptive *'scientific discovery*', as if the author had attempted to patent the wind, palm trees, maybe transcendental numbers, or ammonia salt, on one hand because it is not, by any means, ascribed to nature – very much the contrary, it is a method and architecture of computation, thus fully patentable —, and on the other hand, because there isn't any pre-existence or precedence of matter under any shape or form whatsoever, neither in society, nor in nature.
However clear and indelible the presence of novelty and non-obviousness in the claimant and author's invention may be, what is even more unmistakable is that, even if—purely for the sake of argument—the invention was presumed to be a '*scientific theory*', every aspect of the claims in the international application is definitively intended for practical application. Furthermore, the claims are directly tied to the production of a technical effect, all of which abide to patentable subject matter.
Even when stretching the argument to its widest boundaries, almost to the point of delusion, in an effort to uphold the argument of a '*scientific theory*', one must still recall that if, through isolation or technical ´synthesis' — a pivotal term in the claimant΄s patent –, a reductionist yet essential tenet specifying *'the manner in which the product is produced'* (Sigrid Sterckx; Julian Cockbain; CUP, 2012) were added—while leaving the discovery or '*scientific theory*' itself untouched — then, objectively, both the subject

matter and the statutory category of the invention are patentable. This means that it satisfies the fundamental requirement of patentability, which is usefulness, and falls into at least one, and in truth, all four of the statutory categories described in patent law: machine, manufacture, composition of matter, and process (*Fundamentals of Patenting and Licensing for Scientists and Engineers*, Matthew Y. Ma, World Scientific, 2015)

Yet, more importantly than all of the previous, the invention is, actually, not a '*scientific theory*' by any means, as it shall be clear throughout the reading of all the supporting elements. Any *Opposition* (O) is thus framed, first and foremost, within the domain of patent law and, secondarily, within that of technical matter, although these aspects are often presented together and are ultimately inextricable. This is also the approach taken in this journal paper, even if, for structural clarity, the discussion is divided into distinct sections with respective titles: the now-concluding section, 'Philosophy of Science,' which addresses the ‘demarcation question’, the next section, 'Patent Law,' which examines ‘patentable subject matter’, and the final section, 'Philosophy of Technology,' which explores ‘technical matter’.

The author adds, furthermore, an ending minimalist and compact section of seemingly extraneous matter, but in truth deeply explicative of the fragile stage of knowledge-based philosophy of science, and, by extension, other domains, titled “Law Pseudo-Science”. Indeed, its significance is such that it would be well worth expanding into a critical book on the intersection of philosophy of science & law, specifically addressing the failed scientific demarcation of so-called legal sciences.

Patent Law: patentable subject-matter

The core of patent law, entwined as it is with intangible assets, is grounded in the provision of a potential financial reward, either as an inducement to invent (*ex ante*), or as a means to support the production and industrialization of an accomplished invention (*ex post*). in which case the disclosure of technical information is the lawful necessary counterpart. Despite obvious, encompassing all multifarious forms of history and architectures of patent law, patent prosecution or prosecution history, we shall always call to mind that the goal of patent specification is not, in abstract, pure dissemination of knowledge. Rather, it aligns with the perspective expressed by Craig A. Nard in *The Law of Patents* (Aspen Publishers, 2008): *'To borrow real property terminology, the claims set forth the metes and bounds of the patentee’s proprietary interest'.*

The International application No. PCT/PT2021/050014, ”*Method to Perform Computation at or near the speed of light (typical) digital image-to-binary singular or multiple-nodes/servers and computer architecture*”, in Research & Development (R&D) else coined U-Mentalism (or U.-M. in short), because it does not overlap with prior art, is fully and legitimately set to claim its broadest tenets, provided that it complies with patent law.

A fundamental legal precept is also the idea that, granted that the right to useful inventions belongs to the inventor, ”the public good fully coincides in both cases with the claims of the individuals”, herein ”both cases” (Craig A. Nard, 2008) referring to utilities and copyrights.

This is the reason why, for instance, Article I, Section 8, Clause 8 of the Constitution of the USA, which was passed unanimously, reads – ”*To promote the progress of science and useful arts, by securing for limited times to authors and inventors the exclusive right to their respective writings and discoveries*.” – and also the reason why, despite patent law’s exclusion from common law, there is a manifest public interest in general patent protection, thereby, in the first place, international law having expanded to a *World Intellectual Property Organization* (WIPO), under an *International Patent System Treaty* (PCT).

If all of this is clear, it could not be more certain the assumption that the law— and economics— oriented social science (beyond even wrongful interpretations practice, or misapplication of legal interpretations) and the inherent normative scope and standards of proprietary claims can often prove to be hollow, frothy and very shallow. Indeed, they can do so very quickly, as shown by Brunelleschi’s very first universal conceded patent which, like his boat, sank in the Arno River, thus impelling the more solid *Venetian Patent Statute* (1474), or Jefferson ́s guided American ́s *Patent Act* of 1790, shortly after the Constitution, against excessive litigation, unruly bazaar shopping, and disuniformity. Apropos, a note of the Federal Circuit tells: '*While the role of the claims is to give public notice of the subject matter that is protected, the role of the specification is to teach both what the invention is and how to make and use it.*' (*University of Rochester v. G.D. Searle Co., Inc.*, 358 F.3d 916, 922 n.5, Fed. Cir. 2004), which to this day is truthful under the requirement that the application claims be clear and sufficiently disclose the invention to persons with ordinary skill in the art. But this is not the only thing it is presumed: it is implicitly presumed also that the (social science) interpretative agents, more importantly examiners, possess ordinary skill in the art, and are further capable of understanding the technical disclosure of the invention, without any aggravating deficit or bias.

It has to be taken into account the most possible grievous consequences, if agents, examiners, judges or jury are ill equipped and produce poor judgment. Imagine only if, say, Samuel Morse, after applying (1837), and being granted a patent (1840, reissued in 1846 and 1848) for the “*electromagnetic telegraph*”, had, instead, been denied. Two decades of hard, irksome, and expensive scientific work down the drain. The technical precedence over Steinheil, Cook, Wheatstone, and Davy worth less than a dime. His family heritage, much beyond mere patrimony, yet the proper dear and cherished lives and future steps of his heirs, would have been cast into an alternative, much greyer and far bleaker, history.

Worse still, this would result into a possible wreckage of patent law foundations and a serious devaluation of science, more so if we remember that the inverse happened: *O’Reilly vs Morse* (56 U.S. (15 How.) 62 (1853)) would, as it is known, provide a fundamental clarification on statutory or patent-eligible subject matter, including the very same subject matter of the author and applicant's invention. Indeed, because cycling historical eras in accordance with the philosophy of technology teleology and its inherent arrow of time are major shifting actors in patent law (ethical) teleological shape and law design by legislators, the U.M. control case study might be labelled amongst those whose major sign is, precisely, that of a new General Artificial Intelligence (AGI) period or pre-eon, under which we already live in, that belongs to reasoning based, highly parallel information complexity and self-administrative, highly autonomous, cybernetic wide-general systems. Patent laws swiftly adapted to the electromagnetic era, whilst adaptation to the latter is still put to the test.

Crystal clear is, furthermore, EPC's (*European Patent Convention*, signed 5 October 1973) – a multilateral treaty instituting the *European Patent Organisation* (EPO) providing an autonomous legal system – common standard for patentability. EPC's Article 52(1) reads: "*European patents shall be granted for any inventions, in all fields of technology, provided that they are new, involve an inventive step and are susceptible of industrial application*". This is sustained by overall guidance found in the *Guidelines for Examination*, the *EPC Rules*, and EPO case law from the *Technical Board of Appeal*, under the guise that, when *Rules* conflict with *Articles*, *Articles* override the *Rules*.

Again, in *Exclusions from Patentability, How far has the European Office eroded Boundaries?* (Sigrid Sterckx; Julian Cockbain; Cambridge University Press, 2012) the authors have written: "*Other than to acknowledge that discoveries may lie behind inventions, few Boards of Appeal decisions have addressed the exclusion of discoveries*".

We have, then, to ask ourselves what was the underlying reason behind the *International Searching Authority* (ISA) appointed labelling of "*1a. Scientific Theories*" in the U.M. computer architecture control case study, so slim and scarce it is found the '*scientific discoveries*' labelling itself.

Elementarily, there are only two alternatives in answering this almost unheard-of, and extremely rare to be found labelling of a '*scientific theory*', so rare, in fact, that is its almost of mythical existence, as demonstrated statistically by EPO´s documentation, which corresponds to the portentous moment of the examiners' decision: either we have the derisory, foolish invention alternative; or instead, we have the singularity, non-objectionable invention alternative. Only these two semantic impetuses could have launched EPO´s body of examiners to produce the '*scientific theory*' labelling.

Derisory patents are hammered by patent law, singularity patents are most likely the anvil upon which new patent law is forged. This is a big difference, a due undertaking for EPO, or any other national office of patents' body of examiners to immediately grasp.

In *Patent Law for Computer Scientists, Steps to Protect Computer-Implemented Inventions* (Daniel Closa; Alexa Gardiner; Falk Giemsa; Jörg Machek; Springer-Verlag, 2010) it is wisely noted: "*If you have developed a new mathematical method as an abstract entity, without any concrete implementation leading from that, then your invention will be treated like a new scientific theory or a discovery and you will have no chance of getting a patent. Filing a patent application will then be a quite expensive way to publish your invention, if indeed that is your sole intention. But, if you had to develop a new mathematical method in order to solve a technical problem in a technical* *environment, then you can draft your application in such a way that the technical implementation is completely disclosed. If you follow that path, you then only face the gauntlet of the prior art and other objections the examiner may raise*."

Brilliant! Or, we should say — "Eureka!" — under patent law. This simple regard is the exemplary correspondent to the critical line of Archimedes' law of flotation or buoyancy – "*any body completely or partially submerged in a fluid (gas or liquid) at rest is acted upon by an upward, or buoyant, force, the magnitude of which is equal to the weight of the fluid displaced by the body*" – or, we should rather say a principle of law for patentability – any patent claims and drawings, completely or partially immersed into the legalistic body of patent law, upon examination, are acted by an upward technicality force, the magnitude of which is equal to the weight of the proposed claims and drawings, nevertheless steady and perdurable – thus leaving out open the suggestion to, under the very same regard, read off the difference between a natural discovery and a scientific invention, inasmuch as a natural law and an adopted technological use of any principle of law.

It should be drawn attention to the fact that, inversely related to the episode of *Il Badalone*, Brunelleschi's vessel boat with a patent (1421), yet one such contraption sunk at the first contact with water, there could be the case of a denied patent, whose technicalities are far more valuable than that of a paddled doohickie.

Thus, in our drafted conclusion, under the first alternative—wherein seemingly derisory and foolish claims are put forward, potentially akin to the EPO's classification of a '*scientific theory*' under patent law— we may cite as an example the UK Intellectual Property Office's decision concerning rights in GB0614467.9 and GB0705155.0: Randell Mills' '*Blacklight Power*' (*BL O/076/08*), a patent application on perpetual motion (*WO 2005/067678*).

Another example could be T 1538/05 of 28.8.2006 Yuly Zagyansky's big bang energy perpetual motion applied patent (ECLI: EP:BA: 2006: T153805.20060828), supposedly a fission from Einstein-Bohr physics. This vein is different from more outlandish, trivialised "foolish" tagged patents such as, for instance, by the hand of Arthur Paul Pedrick, "GB1361962A — *Earth orbital bombs as nuclear deterrents*", or alike issued patents for his, or others, similar inventorships. It is also a different vein from, for example, patent court's decision in the matter of the Patent Act (1977), relating to Stephen Thaler's application (GB 1816909.4 and GB 1818161.0) for the AI inventorship of 'DABUS' (*Device for the Autonomous Bootstrapping of Unified Sentience*), programmed to create inventions. As different as these examples might be, the common ground for all is a typification of apparently strongly delusional and eccentric applications – again, characterized here as a bizarre set of claims, coupled with a common perceived serious lack of scientific reasoning and practice –, beforehand the proper assignable labelling of a '*scientific theory'*.

The claimant and author of this paper is, however, very, very cautious to bluntly discredit inventorships, and very gladly engages in a more self-permitted strong scepticism (not to say its Humean strongest form) as he is aware that both in philosophy of science & technology there are plenty paragons of what could be said to be discredited forms of discrediting judgement, in the perfect style of the man hoist with his own petard. A prime case of notice for this take could be the own referred example of (nuclear or not) deterrence bombs. Has not recent interest in nuclear deflection of asteroids suggested that, according to existing models, a near-surface detonation of x-rays would be effective (e.g., Livermore National Laboratory)? In all regards, has not this to be considered scientifically and technologically sound, also dragging patentability criteria with an admissible bias, independently of the clear carrying of significant ethical, legal, and environmental concerns? These concerns, whether or not rooted in international treaties and agreements such as the *Outer Space Treaty* — which prohibits the placement of nuclear weapons in space — remain separate from WIPO's legislation, as long as such prohibitions are not reflected in its body of law, which, in many cases, they are not.

As is proverbially known, beyond the typical criteria for an eligible patent – novelty, inventive step/non-obviousness, useful industrial applicability, as well as eligibility under patentable subject-matter – truth is that this standards oscillate unfirmly between restrictions on patenting inventions related to weaponry or military devices, which may involve national security concerns, and the simple fact that obtaining a patent does not, by any means, imply a legally or purist ethically acceptable use or deployment of the invention.

Has not, for the same case, Elon Musk, the same man that was fiercely taunted, debunked and discredited for his ambition to enter the space industry, thereon proceeding to mark the first vertical landing of a commercial orbital booster with the SpaceX *Falcon 9*, strongly advocated for dropping nuclear weapons on Mars´ poles to evaporate the carbon trapped on the caps' ice? This suggestion cannot, in the slightest, be dismissed as devoid of scientific appraisal and merit, but the starkly inhospitable suggestion, apart from eventual trampling of air and space treaties and conventions, must also overcome the natural suspicion of, for example, Dmitry Rogozin, the head of Russian state space agency, who replied: "*We understand that one thing is hidden behind this demagogy: This is a cover for the launch of nuclear weapons into space*".

There is, in this way, a very wide-open discretionary judgement between the appraisal of Elon Musk´s suggestions and those of Arthur Paul Pedrick, one that remains open to interpretation, including by the EPO's body of examiners. However, in contrast, in the opposite direction of cultural and historical context, their judgement ought to conform to the narrowness and strict legal parameters of patent law slim arbitrariness. An eloquent positive and righteous answer, simultaneously a compelling affirmation of this principle, is given by the fact that Arthur Paul Pedrick's patent was conceded, without the publicity need of "*Nuke Mars*" t-shirts.

For this assessment, let us recall that in the current era, when, under pressure by EU regulation, all new vehicles (cars and vans, as well as the remaining percentage of non-electric trains and airplanes) are set, as of 2035, to be zero-emission, most likely electric and therefore without combustion engines, even if somewhat self-deceptively, how the electric-automobile industry was pioneered by Thomas Davenport (1802-1851), who, together with his wife and a colleague, received the first-ever American patent for an electric machine, titled *'Improvement in Propelling Machinery by Magnetism and Electro-Magnetism'* (*US132A*). Davenport, as early as the 1830s, also developed the first electric car which, although operating on tracks in locomotive fashion, was undoubtedly a prototype for the automobiles that would follow several decades later— such as Gustave Trouvé's *1881 Tricycle Électrique* and Andreas Flocken's *1888 Elektrowagen* — thus launching a paradigm shift with foresight comparable to that of the steam engine era's *Fardier à Vapeur* by Nicolas-Joseph Cugnot. Now, the patent was, at last, conceded, but what is extraordinary is, if we want to compare the legalistic orthogonality and judgemental accuracy of that era and ours, how, despite the motor ostensibly working (a fire occurred, but demonstration witnesses testified positively), a long battle with the U.S. Patent Office ensued, as the U.S. government had never before given a patent on any electrical device, this having been the first direct current (DC) electric motor with an ascribed visible end and purpose.

There is not to find a greater contrast than comparing Thomas Davenport´s death in misery and shortage, with the 1910 ceremony of the *National Electric Light Association* 1910 unveiling of an homage tablet in Vermont — Davenport´s place of birth, life and resting — when the Association´s executive secretary noted out loud to the audience:

"*If the Davenport patent were enforced today, it would embrace every one of the millions of electric motors in use in the United States, whose royalties would constitute an income equal to anything enjoyed by Carnegie or Rockefeller. It would have been a merciful dispensation if the bitter bread of struggle and disaster eaten all the years of his life by this extraordinary genius, this prophetic village blacksmith, could have been sweetened with the merest modicum of the vast wealth that his glowing conceptions have helped create for the benefit of us all*." (Thomas Commerford Martin, former President of the *National Electric Light Association* and of the *American Institute of Electrical Engineers*).

To make it simple to understand, any patent office operating under WIPO and the PCT, say an EPO examiner or the Director of EPO, while by which exact means we do not allude to, but allow ourselves the reserve to freely speculate, furthermore with timely and historical incomparability set apart, are by far wealthier and able to provide for their family generations in the business of inventions and innovations, than Thomas Vermont was (and many, many inventors, today, are), while not risking their time and money, a whole life time, neither having put forward any invention to start with, actually the very first point in order.

Straightforwardly, we are strongly signalling the inevitable downwards curve and decline of bureaucrats in face of scientific and technological sound horizons, alongside self-clinching authoritarian persisted arguments with long years punishment, together with strong proclivity to lobbyism. These are all reasons enough to, once again, advocate for a recommended consultation or a permanent commission that includes the participation of philosophy of science & technology specialists within Intellectual Property (IP) legal and administrative systems and organizations. Nevertheless, the focal point of our attention is, rather, as expressed before, the very high risk inlaid in patent law, by permitting *Search Authorities* (SA) not to produce any *Search Report* (SR) under the excuse of the subsidiary form of WIPO's Rule 39 PCT Subject-Matter, Article 17 (2) (a) (i), i.e., perhaps even premeditatedly deployed, which states that "*No International Searching Authority shall be required to search an international application if, and to the* *extent to which, its subject matter is any of" vi) paragraphs, hereof included "scientific and mathematical discoveries*".

It does not require a lavish imagination to delve into alternate history and imagine only the critical hazard the U.S. patent office could have entailed, in the case of Davenport´s submitted patent.

This is so because it conveyed at the time a strong impetus to attest and classify electromagnetic appliances as nature-given phenomena, attributed to a scientific theory as law-given physics, and to which only the culturally strong industrious American context served as a foil to greater damage. We can even look ahead and convoke U.M.´s control case study in current time, to better precise the diagnosis of the *European Patent Office* (EPO) prismatically ambiguous law interpretation under its very different cultural-historical context and background.

Simply put, because the PCT's "*Declaration of Non-establishement of International Report*", in relation to the claimant´s International Application No PCT/PT2021/050014, declared that no *International Search Report* (ISR) would be established for the sole reason of the subject matter of the application being related to a '*scientific theory*', all that there is to be done, in the eyes of the author and inventor, is to disproof this sole allegation. After a substantial collection of disproved subject matter being presented, demonstrating and showing the '*scientific theory*' hypothesis philosophically, technically and juridically to be hollowed out, not giving the slightest margin to dubious interpretations and devious law- and economics- oriented social science interpretative screeds and quarrels, narratives or disputes, then it is absolutely permitted and reasonable, from the perspective of the claimant, not to dismiss the "funerary box" or "scapegoat" evilly, obscure-intentioned interpretative thesis, except for — again balancing the plates of justice equitably for both inventors and examiners — the hypothesis of derisory or foolish practices of patent examiners, likewise inventors' are found of.

In other words, elementary justice also demands that likewise for the author and, indeed, any inventor in discussions of inventorships, only two alternatives exist, the same be reflected in discussions in respect now to the body of examiner´s *Written Opinions* (WO): either prevails the foolish approach, by result of lack of knowledge or some sort of delusional reverie, or else there is an intentional provoked accident and ill-intentioned strategy, when it is clear that the submitted patent is not, in no event and under no circumstances, a '*scientific theory*'.

It is really best to give the floor and the say to the EPO itself on the topic of "2. Non-inventions under Article 52(2) and (3) EPC" specifically in relation to "2.2 Discoveries, scientific theories and mathematical methods" as in 2.2.1 it instructs us as follows:

"*It was recognised in T 208/84 (OJ 1987, 14) that the fact that the idea or concept underlying the claimed subject-matter resides in a discovery does not necessarily mean that the claimed subject-matter is a discovery "as such*" (G 2/88, OJ 1990, 93)".

Now, knowing that even at the level of the Board of Appeals, very few cases addressed the exclusion of discoveries – and some cases were remitted, as for instance:

- T-877/92 "*not related to a discovery as such*";

- T-869/95 Nissan Motor ”*the claim is not directed to a technical effect constituting a discovery per se, but to a technical instruction concerning a technical apparatus*”;
- T-338/00 ”*Main request - discovery (no) Industrial application (yes)*”;
- T 1213/05 of ”Breast and Ovarian Cancer” strong hold on Rule 29(2)[1]; the similar T-666/05 ”*Main request – Exception to patentability (no)*”;
- T-272/95, citing 'Rule 13ter.1(c)', wherein the EPO states: '*Rule 23(e)(2) EPC made it clear that patent protection should extend to elements isolated from the human body or otherwise produced by means of a technical process, even if the structure of that element was identical to that of a natural element. The claimed H2-relaxin DNA may, thus, be patented, in view of this provision.*'

In continuation, it is for us to discern then, as mentioned in the beginning, the philosophical, technical, and juridical underpinnings of one such taxonomy as that of a ‘*scientific discovery’*, not even obliterating the conjuring of social & individual contextual patterns.

It is appurtenant to this initial frame of analysis, to recall the following:

”*It is well-settled law that, although you cannot patent a discovery, you can patent a useful artefact or process that you were able to devise once you had made your discovery. This is so even where it was perfectly obvious how to devise the artefact or process, once you had made the discovery. The detractors of your patent are not allowed to say: the discovery does not count and the rest was obvious. They are not allowed to dissect your invention in that way. The discovery is an integral and all-important part of your invention. The law does not object to that. It objects only when you try to monopolise your discovery for all purposes i.e. divorced from your new artefact or process. For that would enable you to stifle the creation of further artefacts or processes which you yourself were not able to think of. (UK-CFPH 2005; pars. 9-10)*”.

Apropos the topic of monopolization, no reason to eventually back it was ever adduced, once the author and inventor was always adamant in transpiring that the Intellectual Property (IP) itself, and this is written in the patent text itself, was intended to work as a lawful fortress and stronghold for liberal, open-source, non-monopolistic informatic entrepreneurship, having even stated in the patent text too, that a justly appointed, eventually patentable or not, universal assembly programming language was to be developed under the open-source commercial model, non-patentable as programming languages generally are, but instead apt for OS commercial agreements. This is only one sense of monopoly, having the author and inventor safeguarded himself from unfounded extrapolations.

[1] Rule 29 The human body and its elements: ”*(1) The human body, at the various stages of its formation and development, and the simple discovery of one of its elements, including the sequence or partial sequence of a gene, cannot constitute patentable inventions. (2) An element isolated from the human body or otherwise produced by means of a technical process, including the sequence or partial sequence of a gene, may constitute a patentable invention, even if the structure of that element is identical to that of a natural element. (3) The industrial application of a sequence or a partial sequence of a gene must be disclosed in the patent application*.”
Let us remember that the petitioner ́s invention encompasses bio-computing and quantum computing for that matter.

So, this is the point at which a kind of strenuous Kantian critique, pushed to the extreme of Wittgensteinian aporia, puzzlement or aforethought interrogation surfaces and demands attention. Only doing so, is philosophy of law, and the international law of patents for this case, placed under the trial of the *tribunal of critical reason*. This is not of small importance, as we shall see in the ending section how philosophy of law itself lacks a distinct critical form and era. This deficiency renders the niche of patent law all the more subjugated to erroneous interpretations, most specially in the arena of the presumptive autonomous nature of the law, legal normativity, legal responsibility and sound, verifiable ethical-logical justification.

And the question is very simple, centering on science — namely the *a priori* status of mathematics and the objectivity of law — provoking its limits in regard to the control case study: what does international law of patents dictate, for the sieve of truth and test of reason, if a method of computation were proven within a newly and technically disclosed universal computer architecture? From the appellant's perspective, the answer is equally straightforward, and with this he is answering too to the fundamental dilemma, even if a false dichotomy, for the case in question: to the extent that any sort of computation is performed under the technical guise of the disclosed method of the technical invention or new computer architecture, the proprietary perimeter of the patent ought to be the entire new applicability of the method under the very same disclosed computer architecture technical guise.

The claimant is limiting himself to simply transcribing the patent law teachings laid open for everyone since its inception, which is that in return for disclosing the information necessary for others "skilled in the art" to make the invention, the inventor of the new and useful process is rewarded with a monopoly, usually for 20 years. The patent is, precisely, or shall we exclaim — "Eureka!" — the legal instrument that rightfully and unequivocally protects that monopoly.

"*Anything under the sun that is made by man is patentable*", or so expressed the US Supreme Court in the well-recognised case of an "*invention of a human-made, genetically engineered bacterium capable of breaking down crude oil*" (1980), as long as it is understood that abstract ideas constitute non-patentable subject matter.

For the control case under appraisal, even the argument that researchers and inventors could be curtailed or tactically hampered, does not find any ground for two reasons basically: the first is that the claimant ́s invention is only related to the computer architecture of which there is only singular magnus underpinning, and has no revindications thereof on programming languages or software; and second, because what the invention does is quite the contrary: it definitely fosters and promotes new technical advancements in the computer industry, the shaping of internet, the branches of photonics and quantum computation, and even contributes to the broader civil societal idea of progress through its implementation.

If the cases referred, one such as Diamond v. Chakrabarty, 447 U.S. 303 (1980), in the context of American Supreme Court ratified decision, are considered to belong to a distant past, it is, nevertheless, pertinent to a patent-eligibility test — the 2106 Patent Subject Matter Eligibility [R-10.2019] —, even if under U.S. national context only, pointing out two criteria for subject matter eligibility more at large:

First, the claimed invention must belong to one or fall into any of the four statutory categories. Under 35 U.S.C. § 101, Congress defines these four categories as the appropriate subject matter for any patent: processes, machines, manufactures, and compositions of matter. The latter three categories define '*things*' or '*products*' while the first category defines '*actions*'—i.e., inventions consisting of a series of steps or acts to be performed. Vide 35 U.S.C. 100 (b): (”*The term ‘process’ means process, art, or method, and includes a new use of a known process, machine, manufacture, composition of matter, or material*.”). (See also USPTO MPEP § 2106.03 for detailed information on the four categories).

Second, the U.M. control case study invention must also qualify as patent-eligible subject matter, i.e., the claim must not be directed to a legalistic/judicial exception unless, when considered as a whole, it includes additional limitations that amount to significantly more than the exception itself. “*The judicial exceptions (also called ‘judicially recognized exceptions or, simply, ‘exceptions’) are subject matter that the courts have determined falls outside the four statutory categories of invention, and are limited to abstract ideas, laws of nature and natural phenomena (including products of nature) (…)*”[2] [3].

All in all, even if we´re attaining to the USA Patent Law for now, underlying the well gauged historical and philosophical standard provision therein lied in American soil, eloquently abridged in the US Supreme Court affirmation — ”*Anything under the sun that is made by man is patentable*” — so to oppose, reading from the U.M. control case study as a paradigm for EPO´s historical and philosophical caricatural standpoint – riddled with petty and unnecessary claims, as if discussing “in the good Lord´s name”, or worse, seeking to “take over God´s place” in trying to fill in the role of industry and markets so to let the world know ahead what is or is not a practical effective method, but beyond now, even more unexpectedly as revealed through EPO’s examination, also encroaching upon philosophy of science & technology’s chairs and departments, dreading the appearance and evoking the spectre of a sort of patent dogmatic Thomism — while we must also caution against the inseparable jeopardy and danger inherent in the U.S. administrative state-shaped worship service to business corporations, as the outermost boundary of its legal and economic framework, thus risking to overrun frontiers without adequate (historical-philosophical) legalistic substratum.

However, away from this antinomy, definitely the worse take is EPO´s poor examination on the basis of the excuse of WIPO´s Rule 39 PCT Subject-Matter, Article 17 (2) (a) (i) to avoid conducting any *Search*, therefore evading its fully informed duty and potentially endangering the proper edifice of patent law jurisdiction.

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[2] Alice Corp. Pty. Ltd. v. CLS Bank Int’l, 573 U.S. 208, 216, 110 USPQ2d 1976, 1980 (2014) (citing Ass.’s for Molecular Pathology v. Myriad Genetics, Inc., 569 U.S. 576, 589, 106 USPQ2d 1972, 1979 (2013). See MPEP § 2106.04 for detailed information on the judicial exceptions (Source: USPTO.GOV).

[3] https://ipwatchdog.com/wp-content/uploads/2022/11/EM-Footnotes-2019-Revised-Patent-Subject-Matter-Eligibility-Guidance.pdf

What is, however, crystal clear in the midst of all these considerations, and in summary, is the fact that EPO and USPTO's potentially doubtful practices are all the more shown to be disproportionate in face of U.M.'s control case study, inasmuch as extreme and contradicting provisions thereof found violently collide with the irrefutable patent legal basis of the claimant´s invention.
Refocusing on the U.S. standpoint, let us recall how Conley and Makowski (2004) have pinpointed what is, in truth, an absolute oddity: the fact that, instead of applying law-abiding technical boundaries—and the author is the first to assert that the invention classically conforms to *Diamond v. Chakrabarty*, 447 U.S. 303 (1980), a case about which UNCTAD's Intellectual Property Unit noted: *'Significant is alone whether it is a product of nature or of human invention,'* a principle that aligns perfectly with the U.M. control case study—what has instead pervaded U.S. patent law is the shocking reality that *'patents on genes themselves, as long as they are isolated from the body, are routinely allowed by the United States Patent and Trademark Office and acquiesced in by the courts.'* (*Rethinking the Product of Nature Doctrine as a Barrier to Biotechnology Patents in the United States—and Perhaps Europe as Well*; Conley & Makowski, 2004)."
The result is not far too difficult to guess: ”*The practical consequences of this development are equally startling to the non-patent community: gene patent holders can control not only commercial applications, but virtually all research involving the subject genes. Since genes are valuable as transmitters of information, the ultimate effect can be a monopoly on the use of that information. This, we believe, takes intellectual property protection to an unprecedented level. All challenges to gene patents in the United States have thus far been dismissed almost out of hand*.” (*Rethinking the product of nature doctrine as a barrier to biotechnology patents in the United States—and perhaps Europe as well*; Conley, Makowski; 2004).
It, really, should come as no surprise, as we pointed out in the last paragraph, the telling title – "*Rethinking the product of nature*" – and its post-hyphen addendum– "*and perhaps Europe as well*" of this article. We cannot stand at a point where across the ocean, as in the "Ship of Fools" Platonic tale, patents for modified genes grant the inventor the copyright of nature, while on the other side of the Atlantic, patents for processing methods & new computer architectures are deemed unpatentable, thereof of nature. Shifting sides, it would be equally astonishing if the USPTO were to deny a patent for a computer architecture (except maybe if, of course, the inventor, a person or a corporation, was American) while the EPO´s blockage would derive from the inventor being too European, i.e., much more adhered to philosophy of science above strict applicability, and upholding the universalist and illuminist *Ideen* European tradition.
So much, then, for the supposed impediment to invention halting and the never-stalling rewards for inventors—other than, quite naturally, elite corporations in the biotech industry, as some might argue in relation to gene patents under the USPTO.
Who would not recognize, in this scenario, that the inherent hazard extends beyond biotechnology to all industries—chips and manufacturing, artificial intelligence, super-computation, and more—whether within the USPTO or the EPO? Now, considering the EPO's *Declaration* in the U.M. control case study (even as the USPTO's response remains

pending as of the final writing of this paper, March 2024, revised in February 2025), is this not reason enough for a widespread alert?
Again, *'If something is a product of nature, it should be held to be unpatentable, without ever reaching the questions of its novelty, utility, or non-obviousness'* (Conley & Makowski, 2004) – immediately would have struck anyone - playing the devil's advocate before legislators—to ask whether this is the perfect spot and pretext for the convenient halting and paralysis of an invention?
Isn´t this, in the same step, again the perfect spot and means to forestall and ward off innovation and, in turn, the greatest peril to the patent industry itself?
On the contrary, is it not legally and juridically sound to assess patentability while the *Search Report* (SR) is being produced, without any contradiction? Does the necessary issuance of a *Search Report* (SR) not provide a stronger foundation for an eventual dismissal of non-patentable subject matter?
Isn´t much better not to find any precedence of matter, in which case the exclusivity from nature is far better ascertained from different patent file histories?
Fundamentally, in the most transparent way possible, what the author and inventor is asserting is this: if something is excluded on the grounds of subject matter—such as a so-called *discovery* or *scientific theory*—and deemed *a priori* unpatentable, regardless of novelty, inventive step, or industrial application, then we are faced with a problematic legal mechanism. This norm appears to function as a sort of *magic broom*—riding the industry of natural gene patents while, at the same time, potentially sweeping away inconvenient singularity patents that are otherwise non-objectionable. Consequently, patent law carries an inherent risk of significant jeopardy, especially if national patent offices become ensnared in a law- and economics-oriented social science framework that distorts normative proprietary claims, scope, and standards.
What results is the equivalent to the (annihilation of the) first law of taboo in anthropology, i.e., infanticide or the sacrifice of newborns. Patents are not aborted but let be grown, and once they see the daylight, if singular and even if not at start commercially viable, they are ultimately suppressed, all to appease the goddess of nature.
This is nothing but a lamentable, degrading sign of a benighted, uncivilized society, one clearly unprepared for the liberal, progressive advancement of the value of science.
The absence of civilizational advanced law and practice is all the more seriously flawed, when the overall lasting impression is patent law being generally, and justifiably so, tactically incremental, with minutiae disputed settlements, if not either overly or exclusively so.
An overabundance of legalese and wording imbued with a talismanic aura—constrained within a straitjacket—does not conceal any of these fundamental issues in place of fact-specific inquiries.
It could not be more imperative, then, to emphasize— from a neutral and cultivated critical perspective—the fundamental criteria for patent law eligibility, as exemplified in *Funk Bros. Seed Co. v. Kalo Inoculant Co.*, 333 U.S. 127 (1948) (U.S. Supreme Court):

"He who discovers a hitherto unknown phenomenon of nature has no claim to a monopoly of it which the law recognizes. If there is to be invention from such a discovery, it must come from the application of the law of nature to a new and useful end."

Presumably, since EPO examiners have taken this route in relation to the control case study of the U.M. patent request, what is so far minimally perceived is that they are postulating it prefigures a derisive, foolish inventive case, in either way: either in the case of the invention being sound, valid even under the utmost scientific applicability – in which case the inventor was a fool to recur to a patent with high incurring expenses to present an easy prey to the *'scientific theory'* trapdoor –, either in the case of being an extravagant foolish machination without any grasp to scientific grounds. At the same time and all together, the examiners are saying that the very same provisional control case study patent – "*Method to Perform Computation at or near the speed of light (typical) digital image-to-binary singular or multiple-nodes/servers and computer architecture*", in Research & Development else coined U-Mentalism (or U.-M. in short) – falls within the same line as a discovery, in the sense of something already in existence and effect, like the law of gravity, energy and matter correlation, or else as of a new found gemstone on earth.

In other words, in the eyes of EPO examiners, even if human technical ingenuity could have been demonstrated, it is all negligible under patent law due to the fact that the claimant merely exposed something previously unappreciated in nature. At a time of close remembrance to when the *World Trade Organization* (WTO) and its member nations repeatedly failed an agreement to suspend intellectual property rights for COVID-19 vaccines, thus, refusing to temporarily waive intellectual property (IP) protection on coronavirus vaccines, the claimant is, at best, a vicious robber baron that is preventing a *'scientific discovery'* to be "*free to all men*". Because Art. 52(2)a EPC exclusion is able to be overcome by way of combining a *'scientific discovery'* in a mixed claim with some sort of non-excluded technical method, then the claimant must be really foolish to have sought to patent something in the line of the sun and the planets.

Thus far, we have not even touched upon the *'further information'* in the *Written Opinion of the International Searching Authority* (WO; ISA), where matters become significantly worse—particularly regarding the invocation of PCT Article 17(2)—descending, regrettably, into a fantasy world of unfortunate souvenirs and trinkets.

In the so called patent waste, examiners of patent offices are getting paid, but for none of this appraisal of ludicrous matter – EPO's "*logically untenable*" (Sterckx, Cockbain; 2012) exclusion of discoveries, except, of course, when they most seemingly are such, yet accepted and acclaimed, as with the cited support to EBO ("*European Biotech Directive*": 98/44/EC of the *European Parliament* and of the *Council* on the legal protection of biotechnological inventions) – is the claimant being rewarded for, very much the contrary, losing instead precious time and money.

Indeed, one working seriously in R&D can very easily end impoverished and disenfranchised of contracts, grants and calls, in benefit of a slump of pseudo-science (e.g. collection of gibberish of persecutory cancelling culture, pseudo-historical and falsely purist, instead harassing, defamatory, vulgar and nonsensical revisional agendas, when not spectacular trends of recommender systems and university stars lobbyism) and, in turn, for public amazement, apparently when doing science on their own risk with well-thought investment as the author and inventor´s was, they risk themselves being overthrown of the patent system, now and instead being officially promulgated under a "*Declaration*" that they are effectively doing science, i.e., presenting *'scientific discoveries'*, as much as *'mathematical theories'*.

With all that, unfortunate is he, the researcher or scientist, who must also resort to labour or administrative courts of justice when downtrodden in his rights, and sees formal law being ridden roughshod over by judges whose sense of morality and technical competence is so grievously lacking that even a pitying eye is ultimately unfair.

Gladly, the only thing unfettered by this this unfit framework is that EPO, or for that matter, even the USPTO, need not validate *'scientific theories'* to accommodate for the hijacking new cultural theories and so-called new "scientific" agendas, because university departments and research centres, unfortunately, already do so, and what is more, being given the highest praise and commendations.

For now, the claimant and inventor will simply flag the nonsensical legalese and pseudo-scientific apparatus – the erroneous presumptions underlying the otherwise correct idea that human-made invention must be a result of human ingenuity and inventiveness, never from "revelation" (already a slippery term, as what constitutes nature is habit, and not a series of miracles or revelations, as we read in Spinoza and Hume), in the sense of something naturally occurring, and "*that it must not be a product of nature, whether or not extracted, purified, or isolated and, if related to a product of nature, it must be transformed or altered to become something man-made*." (*Exclusions from Patentability, How Far Has the European Patent Office Eroded Boundaries?* (Sigrid Sterckx; Julian Cockbain; Cambridge University Press, 2012) - for the purpose of emphasizing the EPO examiners' folk idea that either nature is a Matrix-like digital ontology (in the likes of Konrad Zuse, Stephen Wolfram, or Gregory Chaitin), and their own natural ideas and states of mind are akin to computationalism, or else that the claimant has argued in favour of one such theory, when, precisely, what the invention settles is an all *tour de force* against the digital ontology and, beyond that, even more so against non-critically paired digital computation and digital ontology.

Will we reach a point where we would need to come across a miraculous examiner to find any glance of "*technology*", "*machine*" or "*process*" amidst the label of *'scientific or mathematical theory'* in the design of U.M.´s computer architecture?

For the most part, it is true that in patent *Oppositions* (O) and general litigation, questions of technology are determinant in reaching a legally sound decision. But soundness is, in the first place, a logical philosophical-guided concept. In the author´s view, there is an urgent need for philosophical clarification, one as extensive as the latitude of EPO's erroneous presumptions. Since the author does not even know if this conceptual framework took an active role in the intellectual engendering of the EPO's body of examiners, just so to be philosophically clear and legalistically sound, he does not even agree to what Luciano Floridi laid out in *Against Digital Ontology* (2009) – "*A preferable alternative is provided by an informational approach to structural realism, according to which knowledge of the world is knowledge of its structures. The most reasonable ontological commitment turns out to be in favour of an interpretation of* *reality as the totality of structures dynamically interacting with each other*." (Floridi, 2009) –, and has a much more reinforced view of naturalistic induction, where mathematical structures, and even realism are just constructivist naturalistic products of the conjoint human activity and that of nature. Again, to let it be clear, the claimant is much more against digital ontology — or digital (scientific) naturalism as we may call it — than Floridi who critique it, but not so much as to discredit the realism of law and philosophy, so to have clearly demarcated the difference ingrained in WIPO's Rule 39 Subject Matter under Article 17(2)(a)(i) of the Regulations under the PCT. Indeed, the claimant is, through both this article and the official *Opposition* (O), defending the common-sense guided and well-crafted substantive core of Rule 39 attaining to "*subject-matter*", against the unreasonable and illogical interpretation of EPO´s body of examiners as known to this date (February 2025). Yes, indeed, it seems like the claimant seeks to uphold and safeguard the proper enactment of patent law legislation, advocating for a coherent interpretation of exclusions versus the EPO office and its body of examiners' *Declaration*, those who, above all, should be the foremost promoters of such righteous matter.

Stressing to the utmost what is worth paying attention to, on behalf of any patent office, placing oneself in the position of the other, genuinely fostering concern for the serious risk of erring in wrongfully conceding a patent, while safeguarding the requirements of novelty, inventiveness, usefulness, and disclosure, it is due to remember Article 31 of the Vienna Convention – "*General rule of interpretation: 1. A treaty shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose*" (United Nations Convention on the Law of Treaties Signed at Vienna 23 May 1969, Entry into Force: 27 January 1980). This principle is particularly relevant when addressing the recognizable tension between PCT Articles 52 and 53 regarding critical distinctions between technical and moral infringement.

Because PCT Article 53 was not cited in the "*Declaration of Non-Establishment of International Search Report*" in the U.M. control case study, it is all the more easy to fairly settle a simple syllogism test: "*all patent-eligible inventions have technical character; anything with technical character is a patent-eligible invention; thus any example of subject-matter recited in Article 52(2) EPC which has technical character must not be that subject-matter 'as such'* " (*Exclusions from Patentability, How Far Has the European Patent Office Eroded Boundaries?* (Sigrid Sterckx; Julian Cockbain; Cambridge University Press, 2012). This is all straightforwardly statutory, an Aristotelian *Organon*-style easy argumentative dialectic syllogism to follow. No one, really, is reinventing the wheel. Why did not the group of EPO examiners strictly follow the recommendations, the statutory *Rules*, the *Articles* and the *Law*?

As for now, the idea of truth, or the highest possible evidence, when related to the object of science, and the truth needed to be phrased under a plausible articulated formulation, attests that its outcome is beyond weak examination, and constitutes a serious hammering on the incentive to innovate.

Additionally, some contextual notes on the *Patent Cooperation Treaty* (PCT) shall be advanced in this context, insofar it is explicit that the contracting states constituting the union entail that: ”*No provision of this Treaty shall be interpreted as diminishing the rights under the Paris Convention for the Protection of Industrial Property*” (*Patent Cooperation Treaty*), which includes the national laws of a contracting states and their regional national offices. Also to refer is the statement that: *”(2) The objective of the international search is to discover relevant prior art*”; and *”(3) ”International search shall be made on the basis of the claims, with due regard to the description and the drawings*” (*Patent Cooperation Treaty*).

And if any doubts remained, again under Article 33:

”*(1) The objective of the international preliminary examination is to formulate a preliminary and non-binding opinion on the questions whether the claimed invention appears to be novel, to involve an inventive step (to be non-obvious), and to be industrially applicable. (2) For the purposes of the international preliminary examination, a claimed invention shall be considered novel if it is not anticipated by the prior art as defined in the Regulations. (3) For the purposes of the international preliminary examination, a claimed invention shall be considered to involve an inventive step if, having regard to the prior art as defined in the Regulations, it is not, at the prescribed relevant date, obvious to a person skilled in the art. (4) For the purposes of the international preliminary examination, a claimed invention shall be considered industrially applicable if, according to its nature, it can be made or used (in the technological sense) in any kind of industry. ’Industry’ shall be understood in its broadest sense, as in the Paris Convention for the Protection of Industrial Property*.” (*Patent Cooperation Treaty*).

As the author and inventor was informally once told, describing what an examiners’ trainer in a patent national office outside Europe warned about his class of, not before long, patent examiners – “*Do whatever you want, but produce a search report. This is what we are paid to output*.” – by the circumstance of EPO´s *Declaration* in the U.M. control case study being transmitted, it ultimately follows that, even colloquially and proverbially, the simplest logical verdict aligns with fairness in delivered judgments, it is extraordinarily difficult to err so profoundly, such an oddity that it urges being scrutinized and investigated.

Moreover, the only reason for Article 33 not being attended to—aside from *'subject matter'* under the *Regulations*, for which ample justification has already been provided to refute the classification—is that “*the description, the claims, or the drawings are so unclear, or the claims are so inadequately supported by the description, that no meaningful opinion can be formed on the novelty, inventive step (non-obviousness), or industrial applicability of the claimed invention”* (Article 34(4)(a)(ii), *Patent Cooperation Treaty*).

Are we, then, to conclude that the roughly 100 pages of the patent disclosure—submitted with an additional fee to accommodate the extra pages, solely for the examiner’s ease of interpretation and more information —were still insufficient for the EPO examiners’ comprehension?

At this point, it would be unthinkable to also include the extensive *Opposition* (O) document in response to EPO’s *Declaration*—thus adding it to the list of analytically and jurisprudentially unredeemed, poorly apprehended, and demonstrably ineffective forms of empirically exercised (pure) reason by the body of examiners.

With the anchoring of provisions of the *Treaty* prevailing over *Regulations*, PCT Article 16(3) nonetheless establishes the preeminence of the *Assembly Union* (AU) over the *International Searching Authorities* (ISA). In this regard Article 16(3)(c) also states: *"(c) The Regulations prescribe the minimum requirements, particularly as to manpower and documentation, which any Office or organization must satisfy before it can be appointed and must continue to satisfy while it remains appointed.*"

With EPO fully employing approximately 4000 examiners and an annual total of submitted patents reaching 180.000,250 in 2020, this amounts to an average of 45 patents per examiner per year. From the applicant´s perspective, and even more considering the ever-increasing "*special technical features*", particularly across different patenting niches where without much serendipity, there isn't much to awe when one reads headlines such as "*Labor relations turn toxic in the European Patent Office, spike in suicides, staff unhappiness raise questions about work culture at the agency*" (Source: Politico) or "*Employees of the European Patent Office work in a bastion of fear and tyranny*" (Source: Suepo).

Hypothesizing, levelling justice for inventors and patent offices, and for the sake of the argument, any inventor of any invention, but especially, in abstract, of a non-objectionable, singularity-type invention, cannot, in self-explanatory fashion, be the most prejudiced by this hampering and hectic environment. This is all the truer when we acknowledge that the patent business operates in tiny, incremental tactical steps, fuelled by capital injections from large corporations contributing to the endless domino effect of falling and interlocking pieces. Clearly, the risk is a wild shift towards a business model, highly prone to lobbying and elite corporate interests, rather than an exceptional institutional blueprint.

Because Rule 63 of the *Regulations* under the PCT, which states the "*Minimum Requirements for International Preliminary Examining Authorities*" as referred to in Article 32(3) settles that: "*(i) the national Office or intergovernmental organization must have at least 100 full- time employees with sufficient technical qualifications to carry out examinations*", and given that INPI, the National Office in Portugal, which surely does not suffer from labour-related burdens or suicidal tendencies, has suitably cleared the *Search Report* (SR), it is legitimate to think that approximately a more proportionate body of (100 full-employed) examiners, working in a stable environment, are a suitable match and more than enough to handle roughly 100 page patent disclosure. This was, nevertheless, all before INPI, so secure and proud of its report, decided to literally copy-paste EPO´s report, only to later clear it again (February 2024).

The filing of a PCT application is accompanied by the selection of an *International Search Authority* (ISA). The ISA is responsible for conducting a prior art search relevant to the claimed invention, and for preparing the *International Search Report* (ISR) and the *Written Opinion* (WO), before what is called, usually 30 months after the date of the initial application, the "*national phase*".

These guidelines and the associated time period serve as a foundational survey for further examination by national or regional patent offices. The aggravated paucity of arguments and lack of technical prowess and rigour in an unfavourable ISR/WO, is, therefore, a very serious matter, especially given that the *World Intellectual Property Organization* (WIPO), after the release of its *Patent Cooperation Treaty* (PCT) *Yearly Review* for 2020, has clearly shown that international patent strategies are placing increasing demands on the *European Patent Office* (EPO), requiring it to issue 32.2% more International *Search Reports* (ISRs) annually.

At this point, it may be relevant to hear from senior *European Patent Office* (EPO) workers and officials and confront their observance with the only thing with relevance — i.e., the "*subject-matter*" of the invention under appraisal — even more so when their observance is related to computer science, as in the rather articulate book *Patent Law for Computer Scientists, Steps to Protect Computer-Implemented Inventions* (Daniel Closa, Alex Gardiner, Falk Giemsa, Jörg Machek; Springer-Verlag, 2010).

As the gravitational point continues to revolve around *Computer-Implemented Inventions* (CII), with different leverages and varying applications of patent law principles across different Offices (USPTO, JPO, EPO, etc), the methodology for handling such applications is worth studying in the context of the U.M. control case study.

Fundamentally, attention is being drawn to the potentially hazardous effects of the aforementioned law- and economics- oriented social science (possibly extending beyond wrongful interpretations into outright wrongdoing) in relation to the inherent normative claims scope and standards of proprietary claims. This concern arises within the framework of a single report *Written Opinion* (WO) issued by the *International Search Authority* (ISA), especially considering the, even if potentially wrongful and wrongful only, imposed undue uniformity. Once more, should it not be the primary task of the *International Search Authority's* (ISA) to conduct a *Search* for prior art in the first place? What is the point of safeguarding an Aristotelian "formal cause" for the "*subject-matter*" in nomenclature if that very presupposition if grossly flawed? Why call an engineer an engineer if he does not engineer? Why call a *Search Authority* a *Search Authority* (SA), if no *Search* is conducted?

It is both understandable and verifiable that legislators wanted to prevent disasters, thereby making way to the entry of pending patents into different regional or national Offices, even if a prior *International Search Authority* (ISA) and its *Written Opinion* (WO) were beforehand negative by any reason. However, an overly stressed labour culture operating within a one-dimensional personal and corporate mental framework vividly reinforces, again and even more so, concerns about the lack of sound practices, given how pitiful and wretched the signs already are.

Inevitably so and hoping for it to be affirmed, taken that since the 1950s, by pressure from the *Council of Europe* (CE), uniformity was put forward, including the adoption of a standardized classification system, which was later incorporated into the *Paris Convention* of 1971, all together with the establishment of the PCT, and the founding of the *World Intellectual Property Organization* (WIPO) in 1970, and later in the decade WIPO joining the *United Nations* (1974), and the PCT system being officially launched (1978).

This is all regular and expected, as it reflects WIPO's role as a Central Office or the International Bureau for the PCT, even if there is disharmonized congruence, and more so with software-related areas, with one expected result as the following quote tells: ”*The result is an upper limit to all patents that are broadly related to computer-based automation*” (*Patent Law for Computer Scientists, Steps to Protect Computer-Implemented Inventions*, Daniel Closa; Alexa Gardiner; Falk Giemsa; Jörg Machek, Springer-Verlag, 2010).

Nevertheless, patents of significance, and more so patents of non-objectionable (law & technicality), and more crucially, those of singularity type ”*subject-matter*” – obviously not saying that this should be a ”category of classification” –, would not even reach industry or technological field assignment under the *International Patent Classification* (IPC), established by the *Strasbourg Agreement* (negotiated in 1971 and entered into force in 1975) if no type of *Search* was conducted, thus revealed to be truncated, more than denied. This ought to be properly discerned, as the *Strasbourg Agreement* establishes the *International Patent Classification* (IPC) division of technology into eight sections with approximately 80,000 subdivisions, and in case a classification is assigned, a more thorough search for prior state-of-the-art should logically follow. As expressed in official WIPO documents:

”*Classification is indispensable for the retrieval of patent documents in the search for ’prior art’. Such retrieval is needed by patent-issuing authorities, potential inventors, research and development units and others concerned with the application or development of technology*” (WIPO, *Strasbourg Agreement Concerning the International Patent Classification*).

In the end, why is, really, the U.M. patent “*Method to Perform Computation at or near the speed of light (typical) digital image-to-binary singular or multiple-nodes/servers and computer architecture*” (PCT/PT2021/050014) given a “*Classification*”, only to have, then the *Search* blocked? The “*Classification*” given - (G0671/20 *Classification covering simulators which are concerned with the mathematics of computing the existing or anticipated conditions within the real device or system; simulators which demonstrate, by means involving computing, the function of apparatus or of a system, if no provision exists elsewhere; image data processing or generation*) – does not seem to anyone as something provided by nature, as simulators connected to devices or systems, indeed conformed to computation through data processing or generation, is not something to be accounted under natural phenomena. Are patents classified as “*scientific or mathematical theories*” - (Rule 39 PCT Subject-Matter, Article 17 (2) (a) (i), i.e., that “*No International Searching Authority shall be required to search an international application if, and to the extent to which, its subject matter is any of” vi) paragraphs; “(i) scientific and mathematical theories*” – regularly reclassified in reality under the G0671/20 Classification? Does this abstruse error also occur with, say, WIPO´s “*(E) Fixed constructions*”, or “*(H) Electricity*” inventions, suddenly interpreted as natural phenomena? Where is, say, the automotive construction machine taken to be natural phenomena, or that energy-related apparatus deemed a mathematical theory? Has anyone noted, in the meanwhile, that the U.M. patent under review and analysis in the

control case study refers to a system that functions only when implemented within a computing machine operating under a new computer architecture?
And so, in this perplexing tale of a snake biting its own tail, we learn that in order to retrieve ”*prior art*”, the *International Search Authority* (ISA) must first search for ”*prior art*” to classify it, but, as it is manifest, in determined cases where it needs to know what does the technology and identifying the novel method, composition, or process is essential, the ISA effectively declassifies it by not conducting any ”*prior art*” *Search* at all. This is not a symbolic charade by any means, it intends to tell directly that at some point, by this action and governance, the International Patent system will take of his own poison.
Fortunately, both the factual basis and fact-based accuracy still affirm that an invention must fall within the scope of “*patentable subject matter*”, as defined by national law, and (regional office) national law only.
The inventor and author is not suggesting taking the scenic route or the long way around, circumventing the *International Search Authority* (ISA), in which event a massive International Institutional Bureau would be futile and the entire “*international phase*” a total discredit. Quite the contrary. What the author and claimant is affirming is that it be ensured that, once compliant, multilateral treaties and conventions are in place, arising from ratified “*Search procedure*” (SP), which governs the processing of the ”*International Application*” (IA) by the designated *International Searching Authority (*ISA), it would simply be nice and kind enough to adhere to what emanates from Article 15, 18 and Rule 43; 43 bis.1 which explicitly begin with ”*(i) conducting the international search*”. Otherwise, there would be absolutely no prospect of anything ”*preliminary*”, neither ”*examining*”, nor ”*authority*” in the instituting enactment of a ”*Preliminary Examining Authority*”, other than watching the clock go by, as well as an ever-growing accumulation of handling fees borne by the claimant, alongside the necessary and burdensome transmittal of payments to the International Bureau.
It would be worldwide astounding – indeed *technically* unscientific, and *morally* stupid, at best, in one such case of misguidance (or else necessarily judiciously criminal) – to find ahead that EPO’s former 38 national states (including the extended states), along with the *United States Patent and Trademark Office* (USPTO), the *Indian Office of the Controller General of Patents, Designs and Trade Marks* (CGPDTM), the *Japanese Patent Office* (JPO) , and *Canadian Intellectual Property Office* (CIPO), would judge the ”*subject-matter*” of the patent under appraisal as lacking technical character. It would be equally perplexing if they were to state that the invention is not novel, does not involve an inventive step making a (technical) contribution over the state-of-the-art, and that any potential advancement it offers or step ahead, it would be obvious to anyone with a reasonably decent knowledge of the field, and yet cannot absolutely be used, not even as a whole, nor as a part of an industrial process, i.e., in ”*any practical application outside the spheres of purely intellectual or artistic endeavour*” *Patent Law for Computer Scientists, Steps to Protect Computer-Implemented Inventions* (Daniel Closa; Alexa Gardiner; Falk Giemsa; J¨org Machek, Springer-Verlag, 2010).
We have so far reached a point where the fundamental yield of patent law was herein thoughtfully outlined, much like the explanatory statements on technicality referenced

in the official *Opposition* (O) to the *Written Opinion* (WO) in all evidence, which remain fully available for consultation.

Philosophy of Technology: technical matter

What remains to be addressed next is not quite a baseline critical text-to-text confrontation between the *Written Opinion* (WO) and the patent´s technicality core, but rather a broader, free-form critical review, nevertheless retaining at its disposal all the fundamental philosophical, technical-scientific, social-historical, juridical, and patent-law cases considerations related to the direct ”*subject-matter*” as perceived in the heart of the patent's claims and drawings. While these elements are naturally absent here, they remain fully accessible in their own right within the EPO's *Written Opinion* (WO) and the claimant´s *Opposition* (O), both of which are also publicly available in the database maintained by EPO's "*Opposition Division*", yet alluded and all the same herein summarized in encapsulated form.
In response to the *Written Opinion* (WO) notification, the author and inventor submitted an official *Opposition* (O), producing a document of approximately 40 pages, fully detailed with the technical intricacies not only of the invention's implementation but also of its complex interplay with patent law.
Naturally, the full breadth of arguments presented in that document cannot be replicated in this article. However, both the *Opposition* (O) and this article should ideally be read, compared, and consulted in conjunction, as doing so provides a fair and comprehensive assessment of the U.M. control case study.
The International Application No. PCT/PT2021/050014 included from the outset—specifically within the sections *Background Art* and *Technical Problem*, as well as throughout the full length of the patent specification, where it is cited in various contexts—the appropriate technical positioning of prior art, namely *von Neumann's iconoscope memory*.
Moreover, this aligns with a historically significant and highly relevant case: the ENIAC Patent, i.e., the patent on the General-Purpose Electronic Digital Computer by Eckert and Mauchly (U.S. Patent No. 3,120,606).
Since a journal article or academic book is not, strictly speaking, a response to a notification — unlike an *Opposition* (O) which addresses patent law and technical "*subject-matter*" coincidentally — the goal here is not to present a narrative on computer science implementations, but instead to focus only on those aspects relevant to patent law that can elucidate the "*subject-matter*" of the invention under appraisal. For one reason lies in the fact that "*most of the features of the modern computer emerged explicitly in the 19th century*" (Doron Swade, *Origins of Digital Computing: Alan Turing, Charles Babbage, and Ada Lovelace*, 2013), hence seldom suited for patent law, maybe with the exception of the Hollerith patent (US526130A, granted in 1894), and for other reason, the passage and overlapping of electronic computers from

electromechanical devices, which started in the 1930s, was soon affected by interlocking war and military secrecy.
Notwithstanding these factors, there is yet another reason: the fact that the very first patent for an electronic digital computer in history, by John Presper Eckert and John Mauchly (filed 26 June 1947), developers of the ENIAC machine at the University of Pennsylvania is, unfortunately, an illustration of severe judicial malpractice, the courts' non-qualification in technical matters, and anti-trust bias, all of which resulted in a grievous obstruction of justice. It burst into open litigation, and *ab initio* opened the first globally family of lawsuits litigation, which significantly impacted the history of computer science, the fame of which stood under the leading figure of John von Neumann as the author of the final Report.
"*John von Neumann didn't invent the computer, however. The distinction rightly belongs to two men at the University of Pennsylvania, Presper Eckert and John Mauchly. They built ENIAC, the first digital, general-purpose, electronic computer—the first Giant Brain*." (Scott McCartney, *ENIAC, the triumphs and tragedies of the world´s first computer*, 1999), a statement to which the author and inventor of U.M. entirely subscribes. In essence, we have upfront a case, which coincides with the U.M. control case study inasmuch as it is belongs to the same inaugural category of "*computer architectures*" — furthermore the "iconoscope memory" in the so-called von Neumann computer architecture having been raised as "*prior state-of-the-art*" under U.M., besides actual EPO´s *Written Opinion* (WO) heavy prejudice stance toward patent law — on the background of science and technology´s network of engagingly deviated recounted history, all from, again, in the outline, the very same screenplay, i.e., errant patent law practice, but also unqualified courtrooms against the setting of dormant, ineffectively scrutinous, or even absent "skilled in the art" specialists participation.
To this day, the author considers some events related to the ENIAC case eminently anomalous. The author does not quite find odd the expected intervention of von Neumann on the Report, its dissemination and the inevitable association of such a reputed starred personality, more so given his role in the hierarchy of the Manhattan Project, above all because the report itself is brilliantly and concisely written; nor does the author find that odd the lack of hospitability for patent prosecution and securing of rights after Pearl Harbour (1941), considering the mandatory shift in the US economy and its very specialized work force; likewise, it is not quite surprising the need to sale and transmit the patent rights by Eckert and Mauchly to Sperry Rand (first Remington Rand), a large corporation; nor does he find it unexpected in relation to the necessary waiting period after learning that von Neumann´s Report had barred the patent, yet and simultaneously the same date of application (1947), until successfully receiving the US patent (no. 3,120,606O) for the ENIAC (1964); not even so much it is unusual the very unfortunate 1.5 royalty percentage deal (except for IBM), precisely the largest manufacturer by far, who had cross-licensed with Sperry Rand, as strange as it was.
Quite the opposite, what the author finds more abnormal is, after the lawsuit by Sperry Rand against Honeywell Corporation (both Fortune 500 top 100 positions) for patent infringement, the nature of the countersuit filed by Honeywell vs. Sperry with the charge of fraud and monopoly, when, in reality, Honeywell´s filling was minutes earlier, thus,

technically not a countersuit, and difficult to understand as a departing lawsuit; even more so, after U.S. district judge Archie Dawson upheld the validity of Sperry Rand's ENIAC corporate (acquisitions and mergers) patent (1962) vs. Bell Telephone Laboratories "*ideas*", there was the sudden coincidence of having the case jurisdiction based in Minnesota. This followed D.C. Circuit Chief Judge John Sirica's ruling that Honeywell had won the 26th of May race of lawsuit filings by a margin of mere minutes, ironically, in the same state where Honeywell was the largest employer; even more paradoxical was the monopoly anti-trust accusation against Sperry Rand under the Sherman and Clayton Acts via the ENIAC patent, when, in essence, patents have always been about the granting of monopolies, of proof enough the activity of IBM; it draws accrued perplexity the raise of further complications and all the heightened confusion, once, as stated before, these monopolizing accusations towards Sperry Rand — alleging misuse of the ENIAC patent to monopolize the computer industry, stifle competition, and hinder innovation — were as flawed as could be, taken that Sperry had just about commercially alienated its rights to IBM, who would turn out to be, at the end, the target of what Yale law professor Robert Bork called "*the antitrust division's Vietnam*"[4], a constant monopolizing trial target. The supine event of such was the theatrical anti-trust lawsuit vs. IBM which spanned over thirteen years, due to its excessive market power, brought by the U.S. Department of Justice (1969), strangely enough declared "*without merit*" later in 1982, when the Assistant Attorney General William all of the sudden dropped the charges. This decision came less than a decade after the ruling that invalidated Eckert and Mauchly's original ENIAC patent in the case of *Honeywell v. Sperry* (1973)—a time when IBM enjoyed unparalleled dominance, holding more than 70% of the general-purpose digital computer market, reaping deluxe shares and royalties.; and, we have to conclude, all the more abstruse the highly discrediting ruling as invalid of the original ENIAC patent (Honeywell Inc. v. Sperry Rand Corporation , 1973) on the basis of von Neumann´s prior theoretical Report (really of null opposition, given the overcoming of patent barring by mere time past the Report and previous patent grant), and supposedly evidence that Eckert´s duo partner, John Mauchly, had obtained key insights for the design of the ENIAC from the Atanasoff-Berry Computer (ABC) (following Mauchly´s visits to Atanasoff´s house in June 1941), truthful but to the point irrelevant, which altogether severely distracts one only substantive reason only assignable to one indisputable fact, something that the senior district judge Earl Richard Larson gravely failed to juridically and technically recognize: that the ABC computer was, in truth, the first special-purpose, non-Turing complete, non-programmable digital binary electronic computer (with some crucial advancements such as the recapped use of binary, switching and regenerative memory through the use of capacitators), but nowhere near or anything of the sort of the ENIAC, effectively the first programmable, electronic, general-purpose, Turing-complete, digital computer, which convey the fundamental traits that constitute the modern computer. These principles have long served as the foundation upon which the modern computer models its functionality.

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[4] Alec Stapp, The Ghosts of Antitrust Past: Part 2 (IBM), Truth on the Market (February 03, 2020), https://truthonthemarket.com/2020/02/03/the-ghosts-of-antitrust-past-part-2-ibm/

If such a technological and industrious applicability leap cannot support a new patent, then the best is to shut down the entire patent system, and, in all honesty, the same could be replicated in relation to the (lack of) judicial quality as demonstrated by judge Earl Richard Larson, in district courts of law, as jury-type average citizen common sense reasoning seems best tailored to judiciously reveal formal truth.
What´s worse is that the oddities surrounding the ENIAC patent are far from ending here. Beyond this underlying foreign matter — unorthodox and law-unfriendly "anti-trust" bias —, several other instances warrant attention as, for example, the way Honeywell lawyers transmitted to Allegretti's law firm patent lawyer Charles G. Call that there would essentially be no budgetary constraints whatsoever in pursuing leads that could provide information or legal arguments to challenge the ENIAC patent, as if deep-state national interests had met a by-pass through corporations.
Indeed, as a concluding remark, it is strikingly baffling, in relation to the present day, how much the pattern is repeated: unharmful, selectively chosen and preventively discharged, seemingly theatrical, "anti-trust" lawsuits by the US Department of Justice (DJ) and state attorneys.
For example, in the late 1990s, the DOJ case against Microsoft resulted in a final, largely symbolic settlement that abandoned the company's division and monopoly dismantlement. Now, through its partnership with OpenAI—potentially linked via Azure to a highly valued cryptocurrency model—Microsoft enjoys an unrestricted path to AI model commercialization. Similarly, Google has faced numerous lawsuits for monopolizing search advertising markets, dominating digital web advertising, enforcing restrictive practices in the Android ecosystem, and engaging in overall market abuse. However, these cases have largely been softened through settlements and fines significantly lower than the company's profits—including and sometimes especially in Europe—without ever targeting its core surveillance-based advertising business model, which monetizes large-scale user data collection and resale.
This overall market abuse and dulcification with settlements and fines much inferior to profits, based on free-data-recollection-reselling from large user-based, advertising monetization is, indeed, a business model which constitutes a universal snare to individual´s rights, for which we could, for a change, held accountable, instead, the patent industry itself (at least in part, with another part of the blame falling on reckless legislators).
In an era of extreme intensification of the harms caused by the use of data for social and political manipulation through control of the network itself (the internet) as well as the social networks that are part of it, it is bizarre, to say the least, that the original source of this techno-cultural pandemic is not understood, none other than the pointedly flawed data monetization model whose source we are able to identify with precision, nested as it in a patent. It must be strongly emphasized that there has been no greater dulcification and silent sabotage (which would not have gone unnoticed by the political philosophers of the Enlightenment, though it has by far for contemporary ones) than the idea of seizing data that constitutionally belongs to each individual and selling it to commerce, only to resell it—at an even higher price—to the very individuals who were

deprived of it in the first place, and simultaneously, with this process, striping them of several of their prerogatives, civic and libertarian rights in a single stroke.
Now, the patent on which this tragic error is based is US 6,285,999 B1, titled *"Method for Node Ranking in a Linked Database"*, granted to Lawrence Page (co-founder of Google along with Sergey Brin) on September 4, 2001, but assigned to Stanford University through the National Science Foundation, which is to say, as explicitly stated in its text, *"the government"*.
This patent is a prime example of how not only mathematical methods (in this case, statistical methods and basic algorithms) could be said to have been appropriated — something that could already rely on natural theories or mathematical methods, but, beyond that and more important, completely undermining any defence argument by using them in the service of a business model that should be illegal. Once again, if there were any intention to prohibit the application of computational or algorithmic methods based on the exclusivity of natural methods and mathematical theories, then this patent would be the first candidate for invalidation. However, that is not the main issue, as such a prohibition would prevent not only this patent but any patent in this field, including those related to search engine results. The real problem lies in the commercial over-exploitation rights of such methods, which essentially means that someone claims ownership over individual rights that are bypassed through the commercial rights network of the linked database of web pages, rather than just the method itself.
Indeed, Google's empire (as of February 12, 2025, Alphabet Inc., the parent company of Google, has a market capitalization of approximately $2.27 trillion) is built on patents linked to fraud or egregious errors in granting or judicial rulings (its most important foundation IP remains precisely the “PageRank Algorithm”, now the cornerstone of web search and informational civic rights exploitation, not just ranking pages based on their link structures, but also making this information irrelevant due to illegal marketplace of personal database information, additionally, as also previously cited, are also more than questionable all patents related to Google Maps, concerning geolocation, route optimization, and navigation algorithms, some examples including U.S. Patent No. 7,912,753 – *"Determining travel routes using location data, and"* U.S. Patent No. 8,468,778 – *"Systems and methods for updating geographical maps using navigation data”* taken, that, meanwhile, other major Google patents have been notable failures, such as the “Self-Driving Car” (*U.S. Patent No. 8,078,349*), titled *"Transitioning a mixed-mode vehicle to autonomous mode."* and “Google Glass wearable” (*U.S. Patent No. 8,610,730*), titled *"Wearable display device with a tight-fit assembly".*
Therefore, it is no surprise that Google, now Alphabet Inc., has never truly faced an antitrust action with serious consequences, just as it is not surprising that the largest financier ever of the greatest political machine in history, the Democratic Party of the United States, has been, by a significant margin, Google (even if It's important to note that U.S. federal law prohibits direct corporate contributions to federal election campaigns, and these donations represent the collective contributions of individual employees and their families, not the corporations themselves, truth is they are effectively financing with capital originally from Google, and sometimes beyond that, defending directly the interests of the corporation via contributions to federal election

campaigns, the result being, in the recent election cycles, the Democratic Party having secured much greater fundraising in terms of large donations for its presidential candidate, sometimes even twice as much compared with the Republican Party, despite the fact that campaign financing in the U.S. consistently generates strong engagement). Given the significant financial support from individuals associated with Google to the 2024 Democratic Party's campaign, Google, now Alphabet Inc., has not faced serious antitrust actions with substantial consequences. Notably, key members of Google's legal team (Karen Dunn, Jeannie Rhee, and Bill Isaacson from the law firm Paul Weiss), hosted a fundraiser for the Democratic Party during a critical period of the Department of Justice's antitrust case against Google in which they were serving as attorneys.

Furthermore, the Future Forward PAC, which was the largest single-candidate Super PAC in the 2024 United States presidential election, raised over $700 million to support the Democratic Party's campaign, this PAC having received substantial contributions from Silicon Valley donors, including individuals associated with Google. These connections of herein specifically the Democratic Party with Big Tech have for sure influenced its approach to antitrust enforcement, allowing Google to evade significant regulatory challenges, something that had its origin in commercial patent exploitation permissiveness, with shared responsibility of regulatory bodies (something marked by shared blame between Republican and Democratic administrations throughout U.S. history, although worse over time).

The consequences, however, have been the degradation of participatory democracy and a cultural pandemic in the information age, propagated through the model adopted by the so-called social networks. This was all backed by the clear protectionist endorsement of the "*America First*" patent industry and, worse, by legislators with no grasp of the severity of the matter.

The same display of theatrical enactments of antitrust cases happened with Facebook, now Meta, focused on the acquisitions of Instagram and WhatsApp, with the *Federal Trade Commission* (FTC) wishing to force the dismantling of the group, not counting with regulatory and legal compliance challenges as it happened with the cryptocurrency Libra. If considered alongside whistleblowing cases—such as Julian Assange's indictment for conspiracy to commit computer intrusion and violations under the *Espionage Act*, or Edward Snowden's charges of espionage and theft of government property after disclosing classified NSA documents, without any consequences in terms of the idea of judiciously autonomy, moral independence, and protection of the formal status of the law – it evident how there is a misapplied sieve within the U.S. deep-state, where strategic military, global commercial, and international financial interests override the rule of law. This might take the form of an attack from "antitrust" accusations being either reinforced (as with the ENIAC patent), either dismissed (Microsoft, Google, Meta), most probably for the same concealed or deep-state reasons.

Indeed, this also draws a parallel with the major financial crises and great recessions in U.S. history— irrespective of the inflationary debt issuance of the federal dollar—where economic turmoil spills over into society, culture, and science. The 2008 financial crisis serves as a prime example: a gradual erosion of scrutiny at each step of the process led to an unchecked expansion of subprime lending, speculative price bubbles, and the concealment of risks within complex securities. This, in turn, resulted in inevitable defaults and foreclosures, triggering a crisis of public confidence, subsequent bankruptcy failures, and ultimately, pressure on governments and taxpayers to fund large-scale bailouts.

None of this is in the interest of the oldest surviving Federation, the country that has been at the forefront of technological inventiveness, economic growth and political defence of liberalism since its foundation. In all truth, the supreme court´s decision in case of *Diamond v. Chakrabarty* (1980) judge quote from the committee reports alongside the 1952 Act with the information that the Congress intended statutory subject matter to "*include anything under the sun that is made by man*", previously noted by P. J. Federico, the principal draftsman of the 1952 recodification, captures the essence of the Promethean nation, but also its context is remindful of Sophoclean *Ode to Man* passage (Antigone, vv. 332-375) “*He wends his way between the laws of the earth, and the adjured justice of the gods*”, a warning about the cherished indomitable spirit of man.

In like manner we have essayed to bring philosophy of science´s understanding framework to debate firstly, all together non divorceable from the latter is philosophy of technology´s realm of most appurtenant concepts, namely in face of patentability criteria.

We shall start by Martin Heidegger´s (1889-1976) famous essay *The Question Concerning Technology* (1954). The German philosopher settled a severe distinction between two terms in German: *Technik* and *Technē*. The argument goes affirming that, while *Technē* refers to a mode of revealing or bringing forth something into the open, inasmuch common to craftsmanship or artistry, as with the proper metaphysical phenomenology and presented scene of objects, in Husserlian style, on the other hand *Technik* refers to a brute and blunt amalgam of teleological vicissitudes arising from processes and materials. In the case of Heidegger, the advocate of *Dasein*, this distinction carries an important nuance: the case being that *Technik* problematically overlaps with rationality as characterized by instrumentality, efficiency, domination and metaphysical sovereignty over nature. Patentability criteria, as defined in patent law, are necessarily attached to *Technik* rather than *Technē*, in both historical and philosophical contexts.

Nevertheless, it could be mildly disputed that *Technik*, past a turbulent and upheaved appearance, of which the 20th century was the exemplary turnover of “culture” vs “nature”, has metaphysically and phenomenologically set in a deeply ingrained and ever-increasing (machinal) “second nature” turnover vs “culture” itself, in Goethean style. Thus, in a way, *Technik* beyond *Time and Being*, would have inwardly showcased more cosmic, holistic and reflective approaches on truth and human existence, even befriending a sense of awe, with much less dominating and exploitative lineaments in an eventual post-postmodern world, subdued as it is, by technology.

Technik´s hard physicalism, with its strong applicability in the exact sciences, might be reconciling and even reconfiguring the traditional Aristotelian four causes (*materialis*, *formalis*, *finalis*, and *efficiens* in Latin), thereby bringing forth the emergence of *Technē*-like *poiesis*, more concerned with the Greek concept of *physis* instead.

The importance of this reflection goes to the point where it cannot be misunderstood, neither in the negative, nor in the positive, general arguments concerning patentability criteria, i.e., on one hand a largely significant extension of exploitative traits in an invention is not any sign of predilection of monopoly, as it is not, on the other hand, a more holistic and physically-integrated applicability across nearly all domains of knowledge a telling product of metaphysical or natural theory formulation.
Jacques Ellul's (1912-1994) perfectly redirects the tonic on post-modern technology-immersed *Dasein* to the production of values, in between sociology and theology, namely from the idea of a "technological imperative" — reassessed in *The Technological Bluff* (1988) and originally explored in *The Technological Society* (1954), the same year as Heidegger´s work — which offers crucial insights. What is worth rescuing from Ellul´s work to the control case study of U.M., in close attention to the demarcation question of patentability criteria is how the teleonomic logic of efficiency, rationality, and instrumentalism is not only restricted to the agential and actional prolegomena of materials and processes, as it extends also to social, cultural and religious aspects of human existence. Ellul´s portray of technology it of a truly forceful, truncating of freedom and thought, *Technē-Technik* bounded (non-*falsifiable*, epistemically void, over-authoritarian and non- social-liberal) contra-Popperian, erred autonomy.
Naturally, this is not to be understood as some sort of Luddite fury or anti-technological praise, as what it being stated, rather simply, is that *Technē-Technik's dialectic* under *Time and Being*, if not properly and reciprocally antithetic and proportionate, is found to deeply wound humanism, a real danger in face of dulled and darkened *Technē´s-Technik*´s envisagement power of synthesis, now overbearing power of control by *Technik´s* imperative.
National or regional patent offices, including the EPO, can thus be envisaged as both spectrum technological, cultural and social institutions, where technology permeation is instilled through the narrow technicalities applicability of inventions, likewise object´s teleonomy are found of, but also, not just a sort of technological-bureaucratic behaviourism forwardly eroding arbitrium and critic is presumed, in the line of Ellul´s *Propaganda: The Formation of Men's Attitudes* (1962), as there wouldn´t be any higher propagandistic sense of conformity and homogenization than that to be found in leading-edge institutions national patent offices, as if they were the very first victims of technology´s annulments and of what Baudelaire called the "*disease of progress*", depicting the belief on its inevitable ever-ascending affirmative rise, human subjects falling prey to technologized lingo and lorem ipsum. Probably no bigger parody was construed on these premises than Jules Verne´s *Paris in the XXth Century* (1863; first published 1994), a prognosis written at a time imminent to determinant patents (telephone, electric light bulb, typewriter, steam engine and locomotive improvements, the sewing machine, film projection devices and mechanisms, the automobile, the Bessemer process of steel, flying machines and apparel, etc).
Now, philosophy of technology, notwithstanding a modern discipline constituency, lays its underpinnings on ancient philosophy (Plato and Aristotle) with very divergent tides (Rousseauian, Baconian, Kantian, Comtean, Marxist, etc) before its emancipation,

namely after the radical *Scientific conception of the world: The Vienna Circle* (1929) manifesto, edited by Otto Neurath, Rudolf Carnap, and Hans Hahn.

It is, in this aspect, very contradictory the strict verificationist, logical-mathematical analytical reductionism search for a unified positivist, anti-metaphysical, framework, the casting of which took the most active orientation of Neurath himself, with a likely determinist reinforcement, in the post-war period, with the author´s publication of *Empiricism and Sociology* (1950) — where Neurath addresses empiricism and social sciences, defending the use of statistical methods and data to understand social phenomena, grounding sociology as an empirical science, inalienable from observations and evidence — in stark contrast to the perspectivist, non-reductionist, fragmented, postmodern culture in philosophy of technology, underscoring precisely the opposite, i.e., an ever growing attention to social constructive traits of elusive, imprecise and inquisitive stamp.

In the inventor and author's opinion, the crest or signature of post-war philosophy of technology is its targeted focus on social sciences, influenced by critical theory, phenomenology, the political and social dimensions of technology, the shaping of moral and ethical values under the elaborate, inquisitive, and *falsifiable* complexity of *Technē*, precisely at the very same era when patent law, instead, sought for *Technik's* great unification, inasmuch as patent law harmonization efforts culminated in the *European Patent Convention* (EPC) (signed on October 5, 1973, and entered into force on October 7, 1977), and, on an international level, in the *Patent Cooperation Treaty* (PCT) (concluded on June 19, 1970, and subsequently entered into force on January 24, 1978). What is implied herein is that we should not view the latter as an ever-progressive consequence of historical stages — early Ancient acknowledgement of exclusive rights, the Renaissance´s Venetian Statute (1474), the Statute of Monopolies in England (1624), the arrival of Patent Offices at the dawn of the Industrial Revolution, with forerunners such as the *United States Patent and Trademark Office (*USPTO) (1790) and the *British Patent Office* (1852), before the *Paris Convention for the Protection of Industrial Property* (1883) — as if patent law had perfectly armoured itself into a legalese château by the late 1970´s with history in hindsight, after the Industrial and the debut of Information revolutions. Instead, the very opposite is true.

In essence, we should therein read the mounting tension between nature and artificiality, shunned metaphysical *Technē* or else strictly technological *Technik* — respectively, autonomy and teleonomy —, all continuing a crescent "technological imperative" in which nature unfolds itself with each new epoch, in an ever-bounded *Technē-Technik* (less and less *falsifiable*) dialectic. Likewise, in the time period following the *TRIPS Agreement* (1994) – an agreement on *Trade-Related aspects of Intellectual Property Rights* (TRIPS), established as part of the *World Trade Organization* (WTO), to introduce global standards — it is clear how it has raised an ever increasing tension, agitation, indiscretion and series of mistakes, on the background of evident signs: neurotic expansion and receding patentable subject-matter (biotechnology, software, business methods) regarding the scope and limits of patentability; ever new-demanded, yet difficult, patent law harmonization, and notedly, the emergence of patent litigation and law interpretation challenges, with a clear blundered levelness between innovation and monopolies.

At this point, the causes are all together simultaneously the effects.

Again, it is not that post-war philosophy of technology has excessively championed *Technē*, while newly uniformized patent law has upheld *Technik* as a derived cultural-scientific phenomenon. Rather, their persistent and grievous chirality is the most evident sign of a fundamental lack of a central focal point of symmetry, balance, and superimposed equilibrium.

Examples of the latter include the recently introduced *European Unitary Patent* and the *Unified Patent Court*. While the streamlined single-patent system across EU member states promises to reduce costs and administrative burdens, it may also conceal, amid these benefits, an increasingly constrained *Technē-Technik* dialectic—one that erodes critical, systematic, and equanimous falsifiability within *Dasein*, even if there is a hyper-systematic production to hide it.

This remains timely and aligns serendipitously with other relevant *dialectic* signs, such as the streamlining and harmonization of patent prosecution processes worldwide. A notable example is the late-stage PROSUR *Patent Prosecution Highway* (PPH) (2016), involving seven Latin American countries (Argentina, Brazil, Chile, Colombia, Ecuador, Paraguay, and Peru) following their earlier accession to the PCT, and the *Pacific Alliance Intellectual Property Working Group* enhancing cooperation. All in all, these are more or less "organizational culture" examples of unavoidable and progressive harmonization, just as are good retrievable examples under the same lead, the forerunner *Patent Law Treaty* (PLT) (concluded on June 1, 2000, entered into force on April 28, 2005), administered by WIPO, which established a standardized administrative and procedural framework for patent law. Alongside the *European Unitary Patent* and *Unified Patent Court*, the *Patent Prosecution Highway* initiatives, patent law harmonization efforts across the Americas and Asia, and the *Hague Agreement* (2004) further illustrate this ongoing trend.

Nevertheless, as expected and desirable as it might be, with this inevitably comes an increasing risk of hollowed-out lawful legalistic soundness, now transformed into a blunt teleonomic flatness of one *Technik*-imperative, sometimes falling short even of just and fair procedures. well within the confines of *apparent* mundanity aspects, such as application filing, formality requirements, and the overall practices of patent offices in simplifying and streamlining procedures across various jurisdictions.

Again, *Techné-Technik's, falsifiable,* Popperian, social-liberal *dialectic* justly demands and promotes consistency and alignment in patent laws and procedures, with at least a partial *bona fide* intent to facilitate innovation, striving to attest greater legal certainty to inventions, while also fostering global collaboration in IP protection, but the truth is that more and more undermined and non-*falsifiable Technē-Technik* is seen seeping through and frothing up from the just buoyancy principle of patentability criteria.

The control case study of U.M. unparallelly showcases it to the extent that patent law jurisdictional power, inasmuch as innovation worldwide, risks sinking deep likewise Brunelleschi's *Il Badalone* in the river Arno was found to do so, very, very quickly.

Yet again, the focus should be on recommending the promotion of philosophy of science & technology committees and/or advisory bodies within WIPO´s framework for shaping of international IP rules and policies, but more urgent is the need to eliminate the most grievous and, in *dialectical* legalistic terms, *absolute* legalistic *Technē-Technik* non-*falsifiable* pitfall found in WIPO's Rule 39, Article 17 (2) (a), strongly emphasized under the framework of "*(i) scientific and mathematic theories*" permitting that never ever a *Search Report* (SR) by *Search Authorities* (SA) be produced, as shown under the U.M. control case study.

Overall, thus, it is therefore permissible to assert that these leading-edge contextual uniformization signs and trends in the international patent IP system – at their extremes resembling a kind of Marcusean unidimensionality, had the Frankfurt School philosopher not himself fallen prey to core-orthodox dialectic unidimensionality – embody *Technik´s* inevitable teleonomical "imperative". This is a key determinant in Ellul´s assessment of nullifying impossibility, so radical that it emulates the inner arrow of time in a Goethean diagrammatic sense. What follows hereafter is the chiselling away of liberal free, autonomic, and critical exerted Technē, in the image of contextual (progressing or retracting) cultural waves, so to complement Goethe's account of surges, ripples and breakers that oscillate around the, dated as it may be as a conceptualization, inner arrow of time.

Not even anthropological agency or causal efficacy seems, therefore, to be captured alone within the realm of *Technē*, a *falsifiable* liberal-free, ever-meagre and increasingly scant, ethical discretionary *voluntas*, amidst the severe imprint of forceful, teleological *Technik*, within a cultural and contextual *dialectic*, transmundane and transcendental, tides, surges and currents.

Scarce and increasingly pressed as it is, the *Technē-Technik falsifiable dialectic* must be promoted and show evidence of life—achievable only through the complementarity of both terms and their superimposed balanced symmetry. Or, to borrow an image from hard science, it should at least demonstrate its "half-life"—the *Technē*-equivalent in the *Technē-Technik dialectic* — akin to radioactive decay in chemistry. Just as the "technological imperative" of *Technik* under *Time and Being* follows a reaction rate of constancy (now accelerated due to the omnipresence of tech artifacts), the (historical futurist) time required for the equivalent reactant concentration to decrease to half of its initial value is theoretically fractionally shorter and shorter. Thus, in this process, the core of *Technē* becomes increasingly relevant, akin to the original concept of half-life by Rutherford, now adapted to Mumford's philosophy of technology, where the elapsed time—or the number of *Technē's* half-lives—continues to decay within a probabilistic cultural and contextual environment.

Within any cultural theory, it may be said that presenting truisms, often framed under jargon such as "natural challenges" or "intrinsic difficulties" to IP protection, is nothing but an exercise in undisguised mediocrity, serving only as flattened folkloric narrative. These truisms frequently reference issues such as the rapid pace of technological advancements, increasing patent litigation in technology, pharmaceuticals, and biotechnology, licensing through litigation, globalization disputes, irreconcilable ambiguities, dominance assertion, and blocked patent initiatives amidst an overwhelming number of patents per field, which ultimately strain national patent offices' efficacy.

Again, the control case study of U.M. is exemplarily, once by an illustrative case it stresses, by reason of the Rule 39 PCT Subject-Matter, Article 17 (2) (a) "*(i) scientific and mathematical theories*" loophole obviating *Search Reports* (SR), the most flagrant flaw in patentability criteria, philosophy of science demarcation & philosophy of technology critic lines, going at the very heart of the problem in what regards the decay of *Technē-Technik* equanimous *falsifiable dialectic*.

By no means is it intended to fall prey to hypocrisy and not to strike back with cynical scepticism. Rather, it is a deliberate effort to invoke important *substracta* elements, not hiding, in the slightest, the fact that, throughout patent history, seldom or never were *omnia, non partibus* scientific discoveries — wisely so — actually applied for a patenting license. Other way to say it, it is very clear that if there were ever a will to dismiss a non-objectionable, singularity invention submitted for patenting, the perfect garbage bin, funerary box, and scapegoat would be the statistically slim, eccentric, and selectively targeted labelling of a '*scientific discovery*'. Hence, the problematic status and *topus* of Rule 39 PCT Subject-Matter, Article 17(2)(a)(i), concerning "*scientific and mathematical theories*".

Fortunately, justice and the sound legality of patent law find their rescue and balance in a profound forethought, sublimated into a single adage—one that gains even greater relevance amid the difficult contextual interplay between the mundane and transcendental, in a post-human era where the falsifiable *Technē-Technik* liberal-free dialectic is in steady decline. This adage is none other than: *"Anything under the sun that is made by man is patentable"* (*Diamond v. Chakrabarty*, 447 U.S. 303 (1980); S. Rep. No. 1979, 82d Cong., 2d Sess., 5 (1952); H.R. Rep. No. 1923, 82d Cong., 2d Sess., 6 (1952); and Hearings on H.R. 3760 before Subcommittee No. 3 of the House Committee on the Judiciary, 82d Cong., 1st Sess., 37 (1951)). This principle stands as the most authentic expression of USPTO's philosophy of patent law, a firm line of (scientific) demarcation in the philosophy of science, a critical apperception lemma in the philosophy of technology, and, most importantly, the highest embodiment of human endeavour spirit and inventorship bound to humanism — unless it is degraded and rendered a victim of crony capitalism, law´s urban guerrilla, medieval juridical barricading pleadings, or similar afflictions of civilization.

In all truth, this Promethean quotation, as referenced by the U.S. Supreme Court, is the only critical line in scientific, technological, jurisprudential, and cultural terms—the only one worthy of being engraved in stone should a motto for WIPO ever be discovered.

In an effort not to close the philosophy of technology section without the broadest overview, it is essential to acknowledge the many other contributors to the field whose insights and chosen themes help to further summarize and refine the U.M. control case study in greater depth.

A thing of importance is the underlying of the aforementioned in relation to the post-modern dominance of *Technē*-related elusive and indeterminant traits, critical theory's appraisal as philosophy of technology´s most recognizable cultural aura, in pure antagonism with IP protection *Technik*-related background for the same time period, reminding C.P. Snow´s *The Two Cultures* (1959 talk), a violent coup on the subduction of thought and alienization, yet pristine in describing *Technik´*s zeal, diligence, earnestness for details, great pride and also great prejudice. In other words, the over dominant rally of such theme in philosophy of technology can only be a significant sign of equally lopsided and disproportionated stance to overall ideal natural-artificial *Technē-Technik* equated antinomy.

Proof of this is the general proclivity of most emergent authors:

Has Jonas (1903-1993) might be the first name advanced, in the line of a Heideggerian need for a *Technē*-Renaissance, through the assigned "Imperative of Responsibility" (1979), which is also the title of his most impactful book, and draws upon various influences: from natural environmentalism to deep-rooted gnostic and phenomenological eurythmic influences, resonating throughout the history of Western philosophy *Time and Being*. Definitely, what has been called attention in this paper about the decayed *Technē* imperative in the oscillating *Technē-Technik falsifiable dialectic*, conforms with Hans Jonas' legacy of thought.

The French-born interwar-period philosophers Gilbert Simondon (1924–1989) and Paul Virilio (1932–2018) serve as illustrative examples of a new genetic stamp of philosophical inquiry in continental philosophy of technology. In the case of Simondon, it is clear how the growth and development pronouncement *On the Mode of Existence of Technical Objects* (1958), points to a teleonomic autonomy within a new *dialectic*, as if *Technik*'s emergentism had absorbed *Technē*'s autonomic stances — previously assignable only to humanism — into a new life and mode of existence for objects, with humanism itself becoming embedded in a drifting network of designed artifacts. Clearly, "individuation" as a concept is very telling under this guise, as if technical objects were subject to a naturalizing process akin to meiosis within their environmental, industrial, innovative, and cultural milieu.

"Transindividual" is the proper mode of transcendence of objects therein. A new ontology of objects was, thus, recrafted by Simondon, one that, under the scrutiny of patent law, warns about the inextricable *Technē-Technik dialectic* inherent in every individuated invention or new method. This perspective underscores how the naturalizing artificial ontology of any invention is immediately upheld by a Promethean action, agency or diligence against nature as it is found: if something is not found in nature, it is, by definition and mandatorily, not an infringement of "patentable subject-matter" if also novel, non-obvious (or involving an inventive step), useful, and susceptible to industrial or mechanical application – therefore, reprising the dictum: "*Anything under the sun that is made by man is patentable*".

However, within the process of individuation, each object, invention, or process undergoes a contextual evolution — both uniformitarian and morphogenetic — reminiscent of the Cuvier-Hilaire debate on evolutionary theory. This applies equally to individuals and humanism as a whole. Patent offices, therefore, cannot act as divine arbiters or sorcerers determining the fate of innovation. Only inventiveness, the market, industry, serendipity, and, ultimately, *Time and Being* within the realm of humanism can bring them forth.

Paul Virilio, in turn, *Crossing the post-modern divide* — to borrow Borgamnn´s expression — is also profoundly distasteful of professional, when not candidly naïve, technological optimism, claiming for *Technē's*-autonomic stances of reflection to be exercised, now more important than ever due to the new era´s "*dromology*", a sort of acceleration of time that implodes and cracks fissures in the *Dasein* of *Time and Being*, a concept whose coinage comes from the blending of many disciplines, from urbanism to media studies, from philosophy to cinema art, from architecture to weaponry.

For the U.M. control case study, *The Machine Vision* (1988) and *The Great Accelerator* (2010) are of utmost pertinence, as they serve as exemplary guides in philosophical research for critically examining its potential contextual implications. Regarding patent law, it is worth recalling *The Information Bomb* (2000), in press almost in the turn of millennia, a study that shows how information media are weaponized and victim of "*logistics of perception*". It also reveals how postmodern societies are surreptitiously and strongly subjected to a succession of *accidens* under "megastructures", an Aristotelian-Thomist injunction rescued from Medieval ontology, to emphasize the nullified status of *ens*, or *Dasein* alike, among different beings. The core argument is teleonomic in nature: as for instance patent offices, as "megastructures", unavoidably encounter weaponized corporative regimentation, which does not even find any form of antithetic, now impossibly postulated, rugged form of individualism.

This isn´t anymore an analysis of the processual-teleonomic timbre typical of Lewis Mumford´s (1895-1990) critique, warning against the overpraise and overemphasis of mechanization, detailing the referred many times *Technē's* "decay" of humanist values, the downfall of well-being and societal cohesion, as Virilio goes far beyond, exposing technological determinism overpower so that, in characterization, it is looming to an unstoppable verge and debacle.

As previously noted, modern, postmodern and even post post-modern philosophy of technology is seen to have exhibited a profound focus on the socially constructed, elusive, imprecise, and inquisitive aspects of technological development, aiming to formulate a new critique. Yet, it has often resulted in a turmoiled marasmus of viewpoints, lost amidst *Technik´s* compulsory and unavoidable drive. Evident and verifiable signs of these remarks are the paths and philosophical avenues explored by many different thinkers in the field.

Albert Borgmann (1937-present) brought about the "device paradigm", portrayed in *Technology and the Character of Contemporary Life* (1984), as welfare objectification and the objectification of welfare all simultaneously, propelling fast new mechanisms of efficiency and spectacular convenience, presenting a more insidious and pervasive imperative than Ellul's "technological imperative" distanced from a genuine *Technē-Technik dialectic* (one so truthful to itself that it remains verifiably *falsifiable* within the scope of humanism).

Borgmann's framework characterizes *Technik*'s increasingly stringent imperative as manifesting in the objectification of "focal things," leading to a drastic inundation of *ens* and the emergence of an overwhelming new technological culture. This culture, devoid of ethical discernment, ultimately leads to impossible embodiments of even the most fundamental human activities throughout history—such as writing, gardening, walking, or contemplating beauty.

Don Ihde (1934-present) introduces a more hermeneutical and phenomenological stance, where the inner dialogic of the *Technē-Technik´s* free-liberal *dialectic,* in all its mundanity, unfolds between human subjects and interpreters on one side, and technical objects and artifacts on the other. This interplay composes perceptual "embodiment" relations, spanning from *Technē´s* old craftsmanship to *Technik´s* modern digital interfaces, a definite argument for typical technological mediation, and furthest, "*multistability*", i.e., the idea that with this mediation comes a multifarious, fulgurant, abundance of interpretations and viewpoints, really inasmuch the description of an era as the state-of-the-art in philosophy of technology, at this point.

There is nothing to wonder, then, about how the politics of technology entered the scene of analysis in unison, forasmuch men such as Andrew Feenberg (1943-present), Langdon Winner (1944-present), and Bruno Latour (1947-present), have all explored the social-political dimensions of technology.

Andrew Feenberg did this by form of a critique of the expansion of social and power structures growth by technological determinism, highlighting the unfortunate rise of *Technik´s* "instrumental" rationality, faced with the increasingly diminished and perishable "substantive" *Technē's* rationality. Alongside this shift, Feenberg examined the accompanying ever stronger constraints on humanism, in comparison with technical empowerment characteristic of *Technik*, hence framing a post-Heideggerian technological *Gestell*.

Langdon Winner extensively comments on the peril of "autonomous technology" and its social political inherent machinations, the window display having been his analysis of the *Three Mile Island* nuclear accident (1979) in relation to nuclear power and democracy. This reinforces the urgent need for a more interstitially humanist *Technē-Technik dialectic* and a nuanced interplay between their dissimilar forces.

Bruno Latour´s *Actor-Network Theory* (ANT), aesthetically designed as it is, is widely plastic, having partaken in studies on scientific laboratories and, generally, the social construction of science. By affronting the traditional view that scientific knowledge is objectively independent of social influences — a stance that many mistook as a provocation against the neutrality of substantive scientific knowledge — Latour´s quarrelling anthropologically excavated scientific facts to show how they result from complex social and material, individual and contextual networks. Whitin these networks, scientific knowledge is constructed or obstructed through negotiations, controversies, and alliances. Moreover, remembering the U.M. control case study under appraisal and similar cases, Latour's framework can be said to highlight the constrictive grip exerted by business institutions, enterprise-wide corporate governance, and governmental methods. Even more troubling, this influence remains empirically *observable* and *verifiable*, yet seemingly inescapable.

Of course, another naturally expectable outcome of philosophy of technology is the fusion of *Technē´s* ethics-centered analysis with *Technik´s* teleonomy. Philosophers such as Carl Mitcham (1946-present) and Peter-Paul Verbeek (1970-present) represent this tributary line of investigation, or better said, they subscribe to the opposite view, namely that ethics is the primordial source, ontologically distributary instead, within the complex and not easily assessable *Technē´-Technik´s dialectic*. While Carl Mitcham focused on engineering ethics, incorporating ethics as an interdisciplinary framework for shaping urban environments, social protection measures, sustainable ecosystems, and green ecosystems, all in all signatory of a much stronger reflective approach against *Technē's* decay, Peter-Paul Verbeek expands on McLuhan´s original mediation theory, surpassing the notion of ethics as merely an interdisciplinary tool, and retrieving in place of that, a larger ontological status, where a sort of internalized design is made agential and deliberative, therefore ethical, at its strongest acumen, that is, the embodiment of perceptions, values and experience, all in favour of fostering innovation with responsibility across disciplines.

In conclusion, this tour of philosophy of technology aimed to deepen the evaluation of two key-aspects. Firstly, it sought to establish a demarcation line in the philosophy of science and, by extension, in patentability criteria, drawing from the imperatively *falsifiable* and free-liberal *Technē-Technik dialectic* within *Time and Being*, to better assess the U.M. control case study in its broader context. Secondly, it aimed to broaden the perception that U.M. is not only demonstrably within the scope of "patentable subject matter", but also, though more subtly, an invention that embodies philosophical critical reflection, acting as a cybernetic resonating oscillator of a new era within the *Technē-Technik dialectic*. Most definitely, by the fact that, by philosophical critical reflexive thought in philosophy of science, U.M. as an invention yields the *power of synthesis* of a scientific theory is, perhaps, only justifiable reason for such a misassessment and gross mistake by the EPO at this stage (February 2025).

The very first moment the EPO, acting as the partially legislative and fully executive body of WIPO, through its body of examiners assigned the duty of serenely reviewing European patent applications filed by applicants to decide whether to grant a patent for an invention, is called to deliberate, the result is a total misinterpretation or manipulation of *Technē-Technik´s* demarcated lines, strictly in terms of patentability criteria. Beyond this, they have exempted themselves, as a *Search Authority* (SA), from conducting any actual *Search*, precisely at the critical juncture where this procedural formality serves as the indisputable proof of an otherwise impossible assessment. This is the precise knot where we can identify the great risk enshrouded in patent legislation: again, Rule 39 PCT Subject-Matter, Article 17(2)(a)(i), which states that *"No International Searching Authority shall be required to search an international application if, and to the extent to which, its subject matter is any of"* six specified paragraphs, among which we highlight for the case under appraisal: "*(i) scientific and mathematical theories*".

It is here that the great peril of *Technē-Technik*'s imbalance manifests, leading to non-substantive instrumentality and dogmatically paving the way for the legalese lullaby that Langdon Winner called "technological somnambulism." In this way, it becomes the most glaring example of the fall — or *decay* — of a non-*falsifiable*, non-free-liberal *dialectic* on the demarcation question in philosophy of science and technology, as well as in patentability criteria.

If it happens so, and one such Rule 39 is manipulatively used, in a disguised manner, as a "press button" to dismiss preferably non-objectionable, challenging inventions of singularity, the International Patent system, with the extreme poison of its own at the heart, will be ruined, first juridically and executively, then socially and politically.

Law Pseudo-Science:

A third section necessarily needs to be added, in the inventor and author's view, insofar as it is evident — despite professionals and academics in philosophy of science and philosophy of law having been entirely absent from the issue — not due to disregard or omission, but unfortunately due to a fundamental philosophical and scientific critical gap (only explainable through the same referred "decay"): the offspring of patent law-juridical or so-called legal science is, instead, a matter of void pseudo-science.
By the way, this fatal disregard in the philosophy of (law and) science for the primeval, most distributive, and most distinctive case in demarcation — law's unsuccessful establishing as scientific, thus law's pseudo-science status — could even rightfully discourage the author's proposed constitution of a scientific body of consultants or a permanent scientific committee trained in philosophy of science & technology, tasked with furnishing WIPO's constituency with effective agential delegation across national and regional offices and boards of appeal. The author does not, in any way whatsoever, hide that this is an undisguisedly dreadful, yet legitimate and well-founded fear, serving only to show the aggravated scenario of the noted deplorable gap. However, by no means this is dismissive of the general inappropriateness of WIPO´s PCT Rule 39 in relation to mandatory *Search*, our main point of focus.
Indeed, bridging the inherent societal risks — following Ulrich Beck's concept of a "social ontology of risk" — with the demarcation criteria in the philosophy of science, and going far beyond any hypothetical aims on either side of the spectrum — whether astrology, homeopathy, parapsychology, divinatory arts, biblical creationism, and other science-masquerading disciplines; dowsing auras, Freudianism, extra-statistical reductionist Darwinism, and climate change catastrophism; vaccine denialism, testimonial ufology, spiritism, reiki and feng shui, or telekinesis with the dead — it is embarrassingly shameful that law has been entirely disregarded as a subject of scrutiny. In fact, law should be regarded as *mater optima maxima* in the discipline of the philosophy of science and, at the very least, as *prima facie* critical within the philosophy of law regarding demarcation criteria. There is no other pseudo-science that so insistently claims scientific legitimacy while, in reality, being the furthest removed from it. This severe contradiction only exacerbates the many domains of legal practice (criminal, constitutional, international, administrative, commercial, property, family, labour, environmental law, etc.). Given that law remains in a pre-critical state of development,

its intellectual infancy inevitably provokes the single most tremendous negative impact on society, far surpassing all the other aforementioned distortions.

What is worse is that, while scientific associations often call out and counteract pseudo-science — sometimes even through judicial means — the case of pseudo-scientific law is quite the opposite. Beyond its initial critical invisibility, law itself, *ipso forma*, embodies direct legal authority, making it inherently self-biased. Just as a state prosecution cannot prosecute the state itself — hence the existence of parliamentary amendments, legislative proposals, and congressional debates in modern states to prevent revolutions — only from within can philosophy of law and historical subjectivity critically address its pseudo-scientific status. The worst possible outcome, however, is precisely the current *status quo*: legalistic and statutory power, in all its branches, actively reinforcing its own pseudo-scientific nature.

Sure enough, academic studies on contemporary philosophy of law encompass a wide range of important authors and themes, addressing the nature of legal theory, ethics and justice, general jurisprudence, and the normativity of law. They also embark on the conditions of legal validity in relation to prescriptive or constructive interpretative cases acknowledging the inherently evaluative nature of legal theory. They are often well-grounded in the historical frameworks, whether under the rule of law or rule by law, and framed into different theories (Natural, Positive, civil legislative body or common law judiciary Realist, and the amalgam of unsorted Marxist or post-Marxist studies, apocryphally misjudged as “Critical”). But there is none, beforehand, that attends the undisguised pre-critical and pseudo-scientific status of law theory.

The range of themes, covered topics, and philosophies of law authors can be said to be little short of being widely removed from the pseudo-scientific accusations we levy against its core state-of-the-art. The critique implicit herein goes far beyond different “-isms” entrenched wars and skirmishes, and instead targets the philosophical void — *vacuitas legis* — within the philosophy of law itself, as if it has never been substantively replenished, in the simplest said to be short of an equivalent Copernican or Enlightenment revolutions (in physics & philosophy). In more concrete terms, no matter how important and helpful they may be, legal studies of reference are intellectual adornments and corporative embellishments to the appointed pseudo-science status of law. Examples meriting citation, even when constrained to the last century, emerge from a multitude of intellectual traditions and scholarly lineages. Yet, regardless of the selection, the discrepancy remains vast, and these examples are often philosophically unrelated to the *vacuitas scientiae in legis* thesis. This serves as a reminder that the scientific vacuity of law is the casuistic precursor to its subdivisions' *accidens* —such as patent law applicability — exemplified by the EPO’s *Written Opinion* (WO) at present (February 2025). The erroneous evaluation of the U.M. invention as a *‘scientific theory’* is a direct consequence of this fundamental flaw.

For instance, debates against legal positivism and the inherent divisionism of strong methodological empiricism (the rather simplistic take on the presumed existence of a legalist table of universal deduction of clear rules for clear facts), and a juridically robust, interpretative moral integralism, was advanced by Ronald Dworkin ()1931-2013). This discourse is a kindred topic with the broader debate between the scientific method and the refined juridical interpretative moral integralism, even if some of the same ingenuity was carried along with one such thesis as the “one right answer” legal exclusivism, as presented in Dworkin´s *Law´s Empire* (1986). Be that as it may, Dworkin’s work never strays from the moral power of interpretivism, subjecting law to the tribunal of pure reason within the demarcation question. Notably, his *not necessarily Herculean* “one right answer” thesis should, in principle, simplify moral barriers and regimental behaviours, ensuring that barristers and deputies adhere to an objectivist demarcated standard of justice and fairness, making their roles — whether in defense or prosecution — far clearer. Consequently, any deviation from this clarity in cases devoid of discretion or ambiguity would constitute a plain and evident immorality. Instead, what prevails is a widespread mercantilist exploitation of legal proceedings, fostering a perpetually unresolved and dishevelled marketplace of moral (and even penal) indulgences.

A great deal of Dworkin´s insights go against both the analytical and descriptively leaned legal positivism of his predecessor in the Oxford Chair of Jurisprudence, the legal philosopher H. L. A. Hart (1907-1922). His advocacy for an internalist jurisprudential discretionary and recognition-active legalism contra merely normative blind abiding rules under legal realist accounts (such as John´s Austin´s social imposed authoritarianism of law backed by threat or menace), goes in pair also with a sociological reworking of legal positivism toward less rigid forms of enacted law, i.e., a more constructivist framework that admits moral stances as factors of trust or distrust in legalist frameworks. Nevertheless, H. L. Hart´s “persistent questions” never tackled the proper scientific demarcation question in law, which is, definitely, the fundamental principle of prohibition, or the intrinsic taboo within legal theory itself.

Surely, H. L. A. Hart’s interpretivist approach to the epistemological status of law allowed for the emancipation of pure formalism into methodological empiricism without succumbing to a universalist framework of objective and universally applicable principles. His work fundamentally acknowledged the inherent differences between law and the methodologies of the natural sciences. However, the reconciliation of formalism and positivism under legalism should also have embraced a reinterpretation of Kant’s *categorical imperative*—the "kingdom of ends" (*Groundwork of the Metaphysics of Morals*, 4:439)—in pursuit of the furthest *synthetic* and scientifically grounded, ethically just legalism, governed by autonomy, rationality, and, above all, the idea of truth. None of this, however, is found in the work of the author of *The Concept of Law* (1961), who, coincidentally, was a colleague of the pioneering Alan Turing at Bletchley Park.

Karl N. Llewellyn (1893–1962)’s legal realism, with its pragmatic endorsement of legal scholarship and professional legal practice — espoused in *The Bramble Bush: On Our Law and Its Study* (1930) — emphasized vigilance against legal formalism and the pitfalls of legal reasoning. While this could have provided a pretext for a deeper philosophical inquiry into the concept of law, its focus remained firmly on legal arguments "in action" and expertise in societal contexts, both of which are far removed from our *quaestio iuris* regarding the demarcation of science. Nevertheless, any critique that an excessive degree of abstraction burdens the core of legal realism must be rejected, insofar as nothing would more profoundly transform the dialectic of legal actions than a

reassessment of law's pseudo-scientific status, particularly through procedural codes, which represent the right spot where to find legalism "in action".
For example, also under the American context, L. L. Fuller (1902 – 1978) sought to revive natural law theory against Hart´s legal positivism, with the consequence, though, of an impossible neglection of formal engagement with abstract principles to explain the "inner morality of law", in this way already constrictively positivist in the acceptance of legal or social forms of validation.
This was all under a rather difficult exercise of trying to find moralism when devoid of all law´s moral validity, when the detrimental to Fuller´s view sees impossible to find a system without moral ends, precisely to find, after all, Fuller's proper views on legal systems inherently entailing moral commitments, determinant for authority and effective function of judicial systems. Unfortunately, this line of inquiry did not entice the cause of legal naturalism to find the indisputable inner methodology of natural sciences, so determinant in the scientific demarcation question, failing to integrate, in this way, the procedural virtues of science into legal processes and the rule of law in pursuit of fairness and justice.
The case of Leo Strauss (1899 - 1973) is also telling of a general dismissal of the Kantian route of investigation, insomuch as he was, himself, a *sui generis* neo-Kantian (affiliated with the University of Marburg, under the supervision of Ernst Cassirer, and at Freiburg where he received lectures by both Husserl and Heidegger). Only this apparent contradiction can rightly rekindle the lost vein of the Kantian fountain for legal interpretation, taken that Kant's *Metaphysics of Morals* (1797), specifically in the *Doctrine of Right* (*Rechtslehre*), differentiates between "right" (*Recht*) and ethics (*Ethik*), and the universal moral law underpinned on autonomy, freedom and rationality, were rather absent in Strauss´s work. Strauss was highly sceptical of the Enlightenment's emphasis on rationality and autonomy, understood by the Jewish author as a ditch for the rise of historicism and relativism.
Even though Leo Strauss contributed to the call for a return to the classical origins of political and ethical thought (Socrates, Plato, Aristotle), his critique of Kant´s ethics and philosophy of right never served as a departure for a post-Copernican revolution in the reappraisal of law. Nor did it provide a direct counterpoint and was combative against Hegelian historicism, dialecticism, and epistemological relativism (not to misunderstand with the necessarily unravelling historical dialecticism and relativism, as for the rest a matter that justly reinforces the sieve of the demarcation question in philosophy of science). Needless to say, Strauss's work, in its entirety, never directly engaged with the urgent philosophical inquiry into the pseudo-scientific status of law.
Surely, a more representative example of the neo-Kantian route is, by contrast, the Harvard-based and modern liberal thinker John Rawls (1921–2002). Since the publication of his masterwork *A Theory of Justice* (1971), his formulation of the "original position" thought experiment has provided a more equanimous and level conceptual ground, effectively distilling different philosophical traditions into an irreproachable and impartial framework —a conceptual and practical foundation for "justice as fairness" in pursuit of legitimacy and an "overlapping consensus" within the social contract.

Rawl´s method normatively incorporates historical standpoints of political and law philosophy; yet, by employing instrumental rationality—thus operating as a power of synthesis within the domain of *Technē-Technik*'s overarching influence—his framework also possesses the theoretical strength to temper relativistic historicism and unrestrained dialecticism, aligning them with a liberal-progressive account of law.

Generally, the reason why Rawl´s philosophy of law appeals to syntheticism is precisely the historicist undercurrent of philosophical avenues, such as the acumen and depth of Kantianism in its emphasis on the universalizability of the moral individual autonomy; Locke to Rousseau´s influence on social equality and impartiality within social contract theory; Mill´s utilitarianism on societal and, inescapably, the progressive liberal commitment to individual liberty, equality before the law, and democratic governance. Be that as it may, nowhere in Rawls' work do we find a reflection on the pseudo-scientific status of law.

Nor, really, has the demarcation question in law been meaningfully addressed, whether in a more or less sophisticated or palatable manner, by any other author from the wide range of disciplines that have influenced legal theory and practice. Habermas' (1929–) theory of communication, Finnis' (1940–) natural law doctrine, Kuhn's (1922–1996) paradigm shifts, Foucault's (1926–1984) analysis of power and discourse, and Unger's (1947–) theory of the plasticity of legal forms all fail to engage with the fundamental *quaestio iuris* regarding the demarcation of science within foundational legal theory. As a result, the pseudo-scientific status of law remains unexamined, leaving the discipline in a strikingly underdeveloped, puerile, theoretical and practical state.

Even philosophers who have clearly established studies linking philosophy of science to legal studies — such as Larry Laudan in *Truth, Error, and Criminal Law: An Essay in Legal Epistemology* (2006) and *The Law's Flaws: Rethinking Trials and Errors?* (2016) — who have entered the courtroom to investigate the epistemological methods for discerning truth are still nested in a "systems balance risk": the evaluation of proof and evidence in relation to a philosophical preference for either minimizing wrongful convictions or, conversely, avoiding wrongful acquittals.

Beyond the admissibility of "evidence", Laudan's great merit is the tonic he places on the interpretation of reasonable doubt and the questioning of legal proceedings disdain for truth-finding. Notwithstanding, never do we find Laudan as a subscriber to the thesis of legality´s lack of scientific underpinning for the absence of pure reason exercise in law´s interpretation of law, that is, not only the finding of legal proceedings and criminal processes as externalist juridical attuned predicaments, but the proper instituted method by which law is the apriorist internalist relation between knowledge and ethics envisaging the furthest synthetic universal applicability.

Its substituted placement is, thus, the typical customary blend of jurisprudence, philosophy, and political theory, which deviates little from the timely, perennially prevailing, intellectual landscape. Indeed, the same could be said of the Enlightenment period itself, taken that, in terms of strict legal theory and its ascendence of importance, the works of pre-Enlightenment legalists — such as precursor Renaissance and early Modern period representatives of different schools of legal thought — were of outstanding relevance and far surpassed (supposedly the post-Kantian German Idealism pinnacle) Hegelian influence on law's immanent determinateness of civic — and also religious — public or private law.

The examples range from early 14th-century Italian commentators on Roman law, such as Saxoferrato (1313–1357), Legnano (1320–1383), and Ubaldis (1327–1400); Salamanca's 16th-century Thomistic reanalysis of natural and international law, including the rights of indigenous people, by figures such as Francisco Vitoria (1483–1546), Domingo de Soto (1494–1560), and Francisco Suárez (1548–1617); the Dutch school of jurisprudence's influence on international law, sovereignty, and jurisdiction, as seen in the works of Grotius (1583–1645) and van Bynkershoek (1673–1743); and, not least, the irreplaceable contribution of British common law and the adversarial legal system leading up to the 17th-century development of constitutional law—marked by the *Petition of Right* (1628), the *Habeas Corpus Act* (1679), and the *Bill of Rights* (1689)—through the historical influence of legal scholars such as Edward Coke (1552–1634), Thomas Hobbes (1588–1679), John Locke (1632–1704), and William Blackstone (1723–1780).

In essence, the newest tailored concepts in philosophy of law emerged from these foundational contributions, which established the principles that would shape modern legal systems and ultimately pave the way for the Enlightenment, but philosophy of law was never emancipated from nonage, remaining entangled in an uncritical status even after Kant's critical philosophy presented its challenge (even if on the ethical plane and not yet in what would later become *Rechtsphilosophie*, as Hegel wrote) namely, the pseudo-scientific status of law in relation to its proper dialectical method. This pertains specifically to the procedural (civil and criminal) framework for ascertaining truth, which should serve no other purpose than to establish formal truth, following a critical philosophical *petitio principii* — an enhancement that is both a practical moral necessity and a theoretical epistemological imperative.

It is, thus, mandatory for anyone skilled in philosophy of science to overcome firsthand the appalling theoretical chaos of the discipline of (philosophy of) law and its medievally obscure and twisted pixie-dust wizardry in shifting sides and moral, not only in advocacy as a practice of law, in deputies' legislature, and in the judiciary branch, but also, unfortunately, in the most bewildering principles of jurisprudential schools. (All the quoted cases differ in essence, but focusing on advocacy as the lead example, it is outrageous how readily lawyers presume their client's innocence without even firsthand knowledge of the judicial process from the outset, which equates to an open license to lie and profit from this dismal and regrettable behaviour. If law were guided by the same truth-seeking and truth-equivalent minimal standards as sound and well-founded scientific disciplines such as physics or biology, the sieve of truth—established through proof and evidence—would have to be the common ground for both the prosecution and the defense, indeed their only non-adversarial and critical line. Furthermore, severe penalties would have to be imposed on attorneys and counsels themselves if they were demonstrably misleading or untruthful in the process or legal actions).

Indeed, natural law theory upholds rational law as a principle of natural justice, and because the faculty of reason is intrinsically positivist through *Time and Being*, with an adjustable sense of morality bound to law, natural law is necessarily also positivist, just as positivist law is likewise naturalist. This holds true even if there is disagreement on the necessary connection between morality and law, and prior to that, on the *existent, possible,* or *necessary* relation — an equally and strictly logical equivalent *problematic*, *assertoric* or *apodictic* same (critical, judiciary, juridical) relation between morality and law —, within (historical, though not necessarily juridically progressive) *synthetic* time of history and different historical times.

In the same way, positivist law inherently carries natural rationality, which is why, even when dealing solely with statements of fact, there is a presumable assignable moral intentional judgment—i.e., ethics—beyond legal formalism.

But is Hegel not the most prominent author to have established the modern jurisprudential values in relation to both the spheres of natural and positivist law, with a self- and inter-assertive interpretational knowledge base grounded in perceptual experience within history itself — independently of legal formalism and a given epoch's *timely* morality — primarily after *Elements of the Philosophy of Right* (first published in 1821)?

In other words, isn´t the supreme irony that Hegel, whose work revived philosophy of law before contemporaneity, and who, quite literally in an outlaw manner (thus in contravention to a judiciary juridical *Critique*), fully surpassed the pre-established critical limits, through post-Kantian right's conceptual dialectic axiology, is nowhere to be rescued to the philosophical debate, just about the point where he could, at the very least, categorically and *negatively* demonstrate the right drawing of concepts in jurisprudence schools?

Again, this is just about the point where Hegelian lexicon and its philosophical fundaments could help properly demarcate the different philosophy of law schools' predicaments, philosophically and critically (judiciary) ill-founded as they are. Through Hegel´s philosophical necessarily *dialectic* dictionary and lexicon, it could be understood not only how much natural law is inherently positive, and equally how much positive law is inherently natural, but also, *determinatively*, how much, in retrospection, *negativity* (*negativ*, *Negativität*) the so-called "Critical" school of jurisprudence is, in truth, one of exceeded critical boundaries and fatally uncritical, including Hegelianism itself (not to mention how reason fetishist Marxist jurisprudence is in such a way enmeshed in dogmatic dialecticism that its operated inversion has naturally fallen into a full-blown, accelerated *decay* into regression, pre-Enlightenment era, pre-critical state).

Nevertheless, Hegelian theory can, once again, through *negativity* (*negativ*, *Negativität*), bring over a more assertive, comprehensive, and timely range of concepts to philosophy of law, even if originally conceived and later counterfeited within the philosophical system´s errors (in Nietzschean terms, the defects of a "superlative history" theory) (*Beyond Good and Evil*, 1886).

Professional antagonists, nowhere to be found conceiving *synthetic dialecticism* understood as a higher stage of truth (*wahr*, *Wahreit*) akin to the phenomenological spirit (*Geist*) exercising consciousness (*Bewußtein*), would quickly dismiss this reflection as *absolutely* contrary to Hegelian philosophy of law. Quite the contrary, though: in all truth, inwardly, taking its conceptual scheme and absolute (*absolut*, *das Absolute*) adequacy, such eventual assessment is, unfortunately, never reclaimed for a demarcated scientific, and lawful assessment of the values of fairness and justice.

Indisputably, Hegelian *Philosophy of Right* revolutionized philosophy of law, namely through its post-contractualist vindication of the idea of state as the highest and most strenuous embodiment of ethical life, a mediation of normative values for the most advanced *timely* ethical order across all forms of individuated life. It also revolutionized the concept of freedom, understanding it as the realization of reason — a *dialectical synthesis* of rational legal order and the common good — as if evolutionary *synthetic* empirical judgment had preemptively and contemporaneously shaped both ethics and legal theory. It has also proven to attest to the individual rights' naturalistic order, emanating from rational legalist positivism and hermeneutical judiciary interpretation, shaping the newest legal orders and legacies. Most significantly, it provided a comprehensive philosophical framework that *transcended* both the English (common law, parliamentary sovereignty, adversarial court system, stare decisis role of precedent) and German (civil law, legal codes, inquisitorial court system, *Grundgesetz* constitutional framework) legal traditions.

Also, *Philosophy of Right's* directrix has permitted a legitimately reasonable extension towards political affairs, concertedly and in actuality (*wirklich*, *Wirklichkeit*) bringing forth the state, law, and governance. Moreover, Hegelian philosophical lexicon is far richer, as a conceptual toolbox, for enlightening philosophy of law school's premises as it brings about, at one coup, philosophical notions (*Begriffe*) axiology, with transcendentality not precluded. It constitutes the least rigid pallet of glossarist elements, fatally terminological in jurisprudence schools. Encyclopaedically and in all-encompassing form, it is fully demonstrative of their power of representation, even driving towards the tension and constrictive force presumed within the legalist formal *logical* movement of the individual´s self-comprehension amidst the great storm of *dialecticism* (*Dialektik*). It also safeguards the advantage of well-retained rational tensioned form against the contingent and irrational, without prejudice to the feeblest forms of the transient and the dependent, something ostensible through the plethoric use of determinacy (*Bestimmtheit*).

Yet, without any contradiction, and rather as an explanation of the previous point, Hegelian philosophy surpasses the due critical limits of scientific appraisal, intermittent already as they are in the precarious form of synthetic *a priori* judgments of Kantian philosophical hallmark. These judgments are already excessively Newtonian and far removed from the conjectural force of Greek theorems. Greek and Hellenistic *physis* theorems, in and of themselves, rightly constitute the eligible, rightful possession, legitimate, and upright entitled, naturalistic-positive (lawful hyperbolized) only postulates and accrued points of speculative *synthesis*. In philosophy of science, axiomatized theorems thus represent a singularity of critically demarcated, highly evanescent frontiers. Not even Newton or Kant managed to confine their physics and philosophy within the bounds of mere reason, as both remained captive to the concepts of God and morality.

But Hegelian *Philosophy of Right* syntheticism reaches nowhere, even if it philosophically internalizes the cognitive representation — including the distinctive powers of judgment, understanding, and reason — of juridical historicism above English common law and German civil law, if it does not ultimately bring forth judicial reason to finally question the pseudo-scientific status of law.

Fatally and abhorrently, law remains in a state of pseudo-science, while phenomenologically and by its experience (*Erfahrung*) – confronting the court with the laboratory or the observatory — consciousness (*Bewußtein*) and its power of orientation in the guise of science is severely distracted, not fully developed in and for itself (*an und für sich*), thus unmediated through reason (*Vernunft*), and simply hostage of a multitude of immediacies (*Unmittelbare*). In this form, it stands as the furthest away from actualization (*Wirklichkeit*) toward the infinite (*Unendlichkeit*), in the sense of the perfect self-containment of the concepts of ethics, justice, and legal autonomy. This is not to be interpreted in the positive sense, where rationality is rightly undermined and challenged by negativity (*Negativität*), but rather as a fatal flaw, the most grievous of critical errors any science could incur.

Accordingly, law has devolved into mere acritical immediacy states of positivity (*Positivität*), signifying nothing more than the tactical adequacies of equidistance in relation to natural or rational order—things merely existing, or, more realistically, the transient life and eventual expiry of the *status quo*. Likewise the spirit (*Geist*) in men sublates (*aufheben*) the distance between the truth (*Wahrheit*) and correctness (*Richtigkeit*), law fatally fails science (*Wissenschaft*) if it does not warrant speculative reason the means to mirror objective spirit in autonomous legal formality, custom and rights. Law, in this fashion, is object of liability and indictment for *Technik*´s overbearing dominance by futile, cynical display of interests masquerading under the guise of a scientific status aura, although merely a façade.

Now, retracing back to the U.M. control case study more closely, before addressing the question of demarcation of patentability “subject-matter” in patent law, comes first the question of the demarcation of science in regard to law itself. The clear, universally valid judiciary sentencing and judgment, in the “tribunal of pure reason,” to borrow Kant’s expression, delivers the unequivocal verdict that law is blatantly pseudo-scientific.

Law, withholding a similarly *Technē-Technik dialectic*, stands as the most demonstrable, non-*falsified* unscientific example of such a *dialectic* “decay” in the demarcation debate within the philosophy of science. It is, then, highly problematic and quite abusive that, based on a distortion arising from an undemarcated, scientifically unclassified pseudo-science — patent law subsidiary, as all other branches of both public and private law, to the vertex of philosophy of law — a spurious and dismissive “Classification” of a *‘scientific theory’* has been imposed in the U.M. control case study. Precisely the discipline least qualified to define science has, through its subsidiary patent law, wrongly invoked an unpatentable status on the grounds that the invention is, paradoxically, a *‘scientific theory’*. This is not merely a singular flaw by the EPO; rather, it is demonstrably a compounded blunder with verifiable, exponentially damaging effects.

Again: the pseudo-scientific status of law — through subsidiary patent law —, delivers such a tragic pitfall, so contradictory in essence, that it proposes to dismiss, in clear contravention, precisely at the point where it cannot be upheld under the U.M. control case study, in full antagonism, by labelling it a *‘scientific theory’*. Law, the greatest example of a pseudo-science, is that which is invested with the authority to decide on the demarcation of what is and what is not a scientific theory, apostatizing and inveighing against technological progress of science through the libel, in an apparent contradiction, of being a *'scientific theory'*. We think there will be no greater philosophical distortion and vicious circle than the paroxysmal absurdity presented here.

What is more, it is through law´s pseudo-scientific defective status that the most severe and heaviest losses to society arise, impacting the entire span of different fields of *Technē's* human activity due to law's broad influence on social-historical contexts and its general permeability. This assumption, in itself, calls for an entirely all-around much different analysis, one that seeks to identify, within legal theory and practice, fundamental legislation, and procedural acts, blatant and openly undisguised infringements, not to say crime. These constitute serious offenses whether by action or omission, against the legal precedence of law, that is, the very autonomous idea of science and justice, as well as the critical orientation of judiciary reason in relation to general juridical precepts in contemporary societies, yet, astonishingly, apparently exempt of any sort of punishment.

One of the most recognizable tenets of science is, in point of fact, the existence of a scientific method, in general terms the reasoning methods that indisputably contribute to scientific inquiry. Surely, the discussion is not simple, with a strong interplay of deduction, induction, and abduction principles that concur to permanently, inasmuch ratify as dispute, what is understood as knowledge in terms of validity, reliability, causal dependence or explicative relation of observed phenomena. Procedural code in law can, herein and generally, be understood as the method ideally required to be thoroughly imprinted with legal science, by which law judiciously subsidizes the investigation and pursuit of truth, even if only juridical formal truth. More often than not, what is tested in science, faced with the exhaustivity of experimental induction (generalizing from a repeated multitude of specific observations across many different scenarios with maximum variability), is the permanent lawful permission of scientific reasoning itself to frame deductive principles (deriving timely valid, and highly precise specific predictions from general principles). Oftentimes, under these circumstances, a gap arises between induction and deduction, where abductive principles (or inference to the best explanation) come into play, facilitating a judiciously open debate aimed at forming valid, testable, and eventually confirmed hypotheses from observed phenomena.

Whenever there exists a *status quo* of intermittent knowledge-basis with strong abductive principles "in action", the colloquial *duck test* serves as an uncomplicated analogy for the tentative effort of scientific reasoning: "*If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck*". Nevertheless, the immense shortcomings of the infant state of legal philosophy, unfortunately, do not even allow for the overcoming of something as simple as the *duck test*. In both legal philosophy and the practical (civil and criminal) codes of justice, flawed scientific reasoning characterization finds shelter in the "presumption of innocence", even when mechanisms of proof have been exhausted to the point that the only remaining element — impossible to occur unless time were to repeat itself — is the first-hand witnessing of the crime (infraction, infringement, breach, or any offense) by the judges and courts of justice. This is in no way comparable to instances when scientific revision occurs due to the emergence of new paradigms based on new discoveries, prompting the reconsideration of scientific theorizing. In contrast, the context of law is never rooted in the natural world or scientific reasoning (a matter of *physis* and philosophy of science); rather, it remains confined to the banal and mundane process of commonplace adduction of “facts” for the purpose of reaching the best possible, albeit trivialized, court decisions, even when these decisions are integrally challenging and interpretivist in nature.

It is precisely at this juncture that the very concept of ordinary jurisprudence finds its justification, not only within (uncodified, induction-prone) common law but also within (codified, deduction-prone) civil law.

Thus, the *duck test* finds caricaturizing rather than characterization in law, quickly descending into burlesque, whenever the context of judiciary reasoning — of a preclusive nature in relation to the judicial decision — is influenced by external interests such as money or power. This occurs even when legal standards align with conclusive evidence, and inductive formalized proof perfectly collates with deductive formal truth. The pathetic result, under such circumstances, is something akin to: “If an act was investigated as a crime, proof of evidence points to a crime, and prosecution is pursued for a crime, then it is not a crime.”

Remembering the property metaphor in relation to patent law — Craig A. Nard́ s in *The Law of Patents* (Aspen Publishers, 2008): ”*To borrow real property terminology, the claims set forth the metes and bounds of the patentee’s proprietary interest*.”

Now in turn, in legal philosophy, the state of play of philosophy of law is said to resemble feudalism, as law is interpreted as a set of perimeters of interests dominated by *Technik´s* strong, conniving, and harmful vested interests. These often manifest as largely immoral, shrewdly judgemental, late-feudal servitude-style Machiavellian actions under the cover of law’s sovereignty, moreover and in truth, actions that are formally illegal under the right scientific conception and design. Philosophy of law is, in this way, formally detached from philosophy of science, *physis*, or any glimmer of scientific naturalism, even though it shares, at the level of legal production, the *‘superstructure’* (*Überbau*) of its time (in materialist language).

It seems, thus, that law´s state-of-the-art — the holding of the perimeter of land in the metaphor — by rights of attorneys, judges, and deputies, is analogous to lordship and vassalage, where law is not owned outright by individuals. The right of easement, meaning the passage through law´s own perimeter in full, as well as the provision of protection, service, and even sustenance, are made a leverage, in the form of blackmail, as a counterpart to the necessary right to live within duties and obligations.

This is the case and in such a manner that, unlike the public perception of Pompeia (Caesar's wife), legalistic shamanism included – law only has to appear honourable, instead of being so. Dismissed of logocentric methodology, it falls into a “state of nature”, insofar as law survives, platonically speaking, much like real-world naturalism, as irrational, changeable appearances succeeding in imitating the forms of scientific reasoning.

The history of law, when analysed, undoubtedly, has a dialectical *positivity* within and throughout the civilizational trajectory of *Time and Being*. This means that its *positivity* trajectory largely mirrored that of the humanist inner arrow of time: from customary and consuetudinary norms in prehistory and early history to the emergence of ancient legal codes formalizing attainable legal principles with foreseen, more predictable legal actions. The advent of Roman law was particularly significant, shaping both public and private law, alongside a determinant personalist internalist perspective on authority and the court system. The *Corpus Juris Civilis* (Body of Civil Law), compiled under Byzantine Emperor Justinian I in the 6th century, was a landmark in this development. Subsequently, following the compilation of early ecclesiastic laws for different communities, gatherings of Christian faith, canons and synods during the 4th and 5th centuries, the arrival to a state of feudality and Medieval law, with fragmented and shattered apart feudal states and their preferable handled matters of cannon law by church courts, up until the turnover of these feudal dispersed states into rageful nation-states, urging much higher centralized uniform legal systems. Therefore these new monarchies instituted royal codifications on the background of the emergence of common law (*post verbum stare decisis* judicial precedent) in Medieval England, and civil law (*pre verbum* codified statutes) alongside the development of Equity and Courts of Chancery also in Medieval England, while Germanic and other European societies revived Roman law during the Renaissance into modern civil law codes.

The Enlightenment then marked what could be considered the most balanced era of *Technē-Technik* since the Greeks—both eidetic and sensible, both positivist and naturalist. This period introduced a new sense of critical reason, individual rights, and the rule of law as inseparable from the autonomous critical subject, universally emanating from nature. It was during this time that the separation of powers and the concept of natural rights took shape, deeply indebted to critical philosophical advancements.

Yet, a *Critique* is due, as it is legitimate to conclude that law has not progressed beyond the Enlightenment period regarding *Technē*'s "substantive" use of reason—unlike *Technik*'s "instrumental" type of rationality, which has since taken over.

Law has never docked into science, hence having lacked a Copernican Revolution, which all emancipated sciences underwent in their seminal phases, either at that time or ever since.

It is the only pseudo-science in contra-regulatory, regimental form, self-attesting to a scientific aura and an uncontested legislative automatic enactment in demarcation theory within philosophy of science, without, in what regards science and reason, in the actual form, neither the substance nor the end of reason.

The clearest example of this conformity with the aforementioned argument is procedural law (both civil and criminal). It openly encapsulates its methodology within the autonomy of the process itself. Procedural law, enclosed within its own framework and governing the conduct of legal proceedings — defining the methodology, steps, and mechanisms involved — presents itself as if it inherently possessed the "patentability criteria" for the judiciary's conception of justice, the fairness of trials, and the rules of court, all under the guise of prioritizing efficiency in fairness. As if, in essence, the "invention" of justice lay in the *how*, rendering it as lawful as the *why*. Through its historical unfolding, procedural law serves as a revealing indicator of the discipline's dialectical relationship with the (philosophical) concept of truth and of truth's degraded counterpart, the (juridical) concept of proof.

This is as if, by the buoyancy principle of *Technē´s*-shaped autonomous law, akin to substantive reason, the concept of truth had to go through *Technik´s*-moulded very porous, and perforated basins of technical proof, perpetually leaking. *Technik´s*-moulded porous basins — procedural law as it stands — are, in relation to *Technē´s* ethics and justice, a broken mould, from which law, as it is, will never transcend its degraded and wretchedly poor pseudo-scientific status.

In fact, while in any constituted science the concept of truth and proof are siamese, engaging all involved agents in the disclosure of scientific facts through falsifiability criteria — in a court of law, by contrary, the prosecution and defence, in the presence of a jury during plenary sessions, actively, insidiously and blatantly deceive and lie to the extent that the system allows, a phenomenon not found in any other constituted science.

The same applies to scientific production in comparison to legislative processes — while proper scientific research methodologies, whether conducted in social constructed milieus as laboratories or, times before, even gatherings of literate men in coffee houses or households — far supersede the compared value of legislative output, which is often tampered with, rigged, obstructed, or promoted by attorney offices, deputies in parliamentary commissions, or any court of justice. Yet, in terms of the ascertained *public* value of a constituted science, and even more so due to the feudalist property tenure within the perimeter of action of law professionals, the inherently fetishised advocacy of inflated customs, excises, fees, rates, taxes and tolls surrounding legislative, advocative, and juridical uses of rules and governing, also artificially inflates the social value of law.

Very clearly, phenomena like incompatibility rulings and perjury in court are socially constructed softened bans, as for the rest never fulfilled or executed, under a sense of Foucauldian "social uses" regimental distancing from penalties, a pure exoneration, if not a pious expiation. Likewise, we find "performance acting" by presumedly professional legal regulatory bodies inspecting these cases, where the severity of the offense, if ever acknowledged, is ultimately parodied into protective corporatist suspensions or, rarely, disbarments.

Against any other eventuality, it is not possible to open any judicial case in which law itself is accused of unlawfulness by demarcated science, nor can legislators be prosecuted, once law´s official duty, in itself, entails automatic control and immunity. For this reason, only a wise legislative turn can positively transform jurisprudence and the practice of law.

Because this section serves as a pure addendum, we are limiting our arguments to this brief exposition, in support of the U.M. control case study. Nevertheless, it remains incomplete at this stage if the most paroxysmal unscientific and manipulative take is not brought to light — namely, how far lawyers, attorneys, solicitors, barristers, legal practitioners, or any legal professionals such as judges and prosecutors, and even law

theorists, are able and willing to go in severely distorting the truth by falsely equating law with well-grounded science.
Because subsidiary patent law, by *negativity* (*negativ*, *Negativität*) discerns the rationale behind patentable "*subject-matter*" as opposed to *'scientific theories'*, it is praiseworthy to accentuate the flagrant extreme contravention of sound philosophy of science demarcation critical line principles arising from law theory and practice.
Even if ending this journal paper with a parody of informal guise — an anecdotical mark if it weren´t so serious — it could not conclude without invoking the para-scientific (colloquially) heard-of claim that deserves to be considered the most twisting perverse jurisprudential argument and, simultaneously, the most flagrantly outlandish naturalist flaw. This claim, as if it was the presented "*subject-matter*" for a patentable claim in law´s admission into the group of sciences, asserts that law itself, in a supposed clincher argument, shares an indeterminacy akin to that of quantum mechanics, literally the farthest scientific knowledge has advanced as a prevailing and overwhelmingly successful predictive fundamental theory.
Just to note, it is worth dissecting this phantasmagoria: the claim suggests that law is allowed to use (indeterminately, without moral and fairness determinacy, a hideous bypass of randomness and statistics by *Technik's* manipulative reason) both truth and falsity bivalently against their most formal truthful instantiation — determined by which side is paying the legal fees in the case of lawyers, and often so in the case of deputies and even judges. This is supposedly justified by drawing a parallel to quantum mechanics, where the necessary physical incompleteness of the pre-measured and pre-observed probability wave distribution is taken as an equivalent to law's indeterminate nature — allowing it to obscure, rather than clarify, objective determinations such as whether offense or crime X was perpetrated by subject A.
This abstrusity is meant to unveil the supposed ontological equivalence between two vastly different situations: a particle accelerator and a judicial court. The first scenario involves a multi-billion-dollar, state-of-the-art high-energy physics collider, a marvel of civil engineering equipped with meticulously arranged technological materials — tons of steel for magnets, vacuum tube systems, miles of superconducting cables, and highly precise detectors, in addition to radio frequency accelerators spanning its length. Its construction, broken down into multiple stages, rivals that of the aerospace industry. It is operated by a vast network of tens of thousands of active collaborators from around the world, including physicists and administrative clerks, a collective effort comparable in scale to an entire city. The singular goal of this immense undertaking is to accelerate protons and neutrons — each approximately 10^{-15} meters (1 femtometre) in size — as well as point-like, nearly immeasurable electrons to nearly the speed of light, forcing collisions under conditions dominated by quantum effects such as wave-particle duality, uncertainty, and non-local entanglement. This represents the most exhaustively constrained and scrutinized experiment in the history of science, yet also the most predictively successful. Its purpose is to achieve determinacy in probability through the wave function, a mathematical description of a system's quantum state, where the square of the wave function's magnitude at any given point determines the probability density at that location. Upon measurement, the wave function collapses into a definite

state, with each instance yielding a deterministic prediction based on initial probabilistic conditions. This is, in sum, a painstakingly ordered human endeavour, where the natural world is exhaustively restructured against an environment of near-vacuum.

This first situation is presumed to be ontologically equivalent to the largely legalistic chatter of lawyers dressed in suits on a courtroom dais, whose primary concern is not the establishment of truth but, rather (except when coincidental), their client's interests and the fees agreed upon in their contract. Meanwhile, this purportedly scientific framework of law can, without a doubt, be tested under any repeated formal assessment of the actions of deputies, representatives, assembly members, and, moreover, judges, court arbiters, advisers, counsellors, and prosecutors. This demonstration of injudiciousness and shallowness in argumentation ultimately settles the question of modern legal philosophy's inability to transcend its intellectual and aesthetic ethical confines.

By this line of unreasonable and totally amiss arguments in favour of law being scientific, it is hardly surprising the labelling of *'scientific theory'* of the U.M. invention by EPO examiners, bureaucratically blinded in their observance of patent law, up until this moment in time (February 2025). Even considering that formal legality is, in some ways, estranged from ethics and justice — particularly evident in the flaw of Rule 39 PCT Subject-Matter, Article 17 (2) (a) (i) — alongside a general cultural disregard for philosophy of science and technology, U.M. ("*Method to Perform Computation at or near the Speed of Light (Typical) Digital Image-to-Binary Singular or Multiple-Nodes/Servers and Computer Architectur*," PCT/PT2021/050014) should never have been legally dethroned as a legitimate and lawful patent. It is clearer than anything now, (patent) law in ipseity being the case here, to which extent it is juridically proven (inasmuch as judiciously sound) – as demonstrated both in this very same journal paper and in the consultable *Opposition* (O) to the *Written Opinion* (WO) documents —, in substantive reason, the inadmissible and manifestly unfounded, prone to immediate rejection, EPO´s *Written Opinion* (WO) on the basis of U.M. being a *'scientific theory'* (as of the time at the end of writing of this paper, February 2025), in this way unwarrantedly and illegally obstructing the patent for no objective, valid, or legal reason.

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